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Sequential Fractionation of Heavy Metals in Soils around the Iron Smelting Factory in Mafraq, Jordan

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A thesis Submitted in Partial Fulfillment Requirements for Master Degree of Science in Department of Chemistry, Yarmouk University, Irbid-Jordan.

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Dedication

This work is dedicated for

My family

My Friends

And

For Whom I Love

Acknowledgment

First and for most I should offer my thanks obedience and gratitude to Allah the great whom I receive guide and help.

I wish to extend my appreciation to my advisor Professor Dr. Idrees Al-Momani for his guidance, fruitful discussions, continuous support and encouragement through the course of this work.

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List of Abbreviations

AAS	Atomic Absorption Spectroscopy
ANOVA	Analysis of Variance
ATSDR	Agency for Toxic Substances and Disease Registry
BCR	Community Bureau of Reference
US-EPA	United State- Environmental Protection Agency
DL	Detection Limit
FAAS	Flame Atomic Absorption Spectroscopy
IC	Ion Chromatography
ICP-OES	Inductively Coupled Plasma Optical Emission
NIST	National Institute of Standard and Technology
RSD	Relative Standard Deviation
SRM	Standard Reference Material

Abstract

Al-Shradqah, Emad, Khaled: Sequential Fractionation of Heavy Metals in Soils Around the Iron Smelting Factory In Mafraq, Jordan.
Master of Science Thesis, Department of Chemistry, Yarmouk University, 2015, (Supervisor: Prof-Dr. Idrees Al-Momani).

This study aims to estimate the levels of selected heavy metals (Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn) in soils around one of the smelting factories in Jordan (Chinese Iron Smelting Factory (CISF)) in Mafraq. Thirty two Surface soil samples were collected from different locations inside and outside the factory during April / 2014 and August / 2015,at 0-15 cm soil depth. In addition, ash samples were collected behind the plant chimneys. All samples were sieved by (250 µm) and then totally digested by strong acids (US-EPA 3052). Some of the collected samples are also sequentially extracted by using the standard BCR procedure. All samples were analyzed by Flame Atomic Absorption Spectrometry (FAAS) for; Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn.

Average concentrations of measured elements were; (2.72 ± 3.72, 98.83 ± 18.99, 35.29 ± 11.04, 45.58 ± 59.62, 19073 ± 4301, 867 ± 240, 87.9 ± 22.6, 500 ± 251 and 514 ± 407) mg/Kg, for; Cd, Co, Cr, Cu, Fe,

Mn, Ni, Pb and Zn, respectively in soil samples from inside the factory. While average concentration of Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn in the outside soil of the factory were; (2.58 ± 4.03, 78.26 ± 32.03, 47.54 ± 33.18, 31.88 ± 4.24, 14516 ± 5539, 1142 ± 398, 271 ± 172, 88.6 ± 27.7 and 194 ± 75) mg/kg, respectively.

For the sedimentary ash the average concentrations of measured elements were; 309.74 ± 155.6, 111.1 ± 32.42, 109.18 ± 67.71, 382.06 ± 161.32, 22220 ± 2079, 2986 ± 958, 153.7 ± 85.12, 7509 ± 1540 and 2955 ± 132) mg/kg for Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn, respectively.

Results of speciation study indicated that, significant fractions of measured elements are present in the bio-available fraction. The highest fractions in the bio-available fraction were observed for Cd, Pb and Co, by nearly 49%, 32% and 16%, respectively. However, most of the Cr, Fe, Cu and Zn were tied up with the residual fraction (93%, 88%, 75% and 55%), respectively.

Key Words: Heavy metals, Chinese Iron Smelting Factory, FAAS, Sequential Extraction, Bioavailability

Chapter 1

Introduction

1.1. General view

Heavy metals are naturally occurring elements that have a high atomic weight and a density at least 5 times greater than that of water (i.e. $D \geq 5$ g/ml). Since the toxicity is the major concern in most studies, the term 'heavy metals' include also the toxic metalloids, such as arsenic, that are able to induce toxicity at low level of exposure. In recent years, there has been an increasing ecological and global public health concern associated with environmental contamination by these metals. Some element are highly toxic, and therefore, considered as the priority metals. Examples of these are; Pb, Zn, Cd, Cu, Cr, Hg and As (Tchounwou, 2012).

All elements are present in the earth crust at different concentrations. Some elements are classified as major constituents (such as, Si, Fe and Al) and some others are present in minor quantities. Elements in nature present in different chemical forms, but they are all usually contained in hard matrixes that are difficult to breakdown. Recently, human activities have dramatically increased, and the need for the extraction of metals from their natural minerals has extensively becomes greater than before. The widespread use of different metals in so many human activities such as

mining, metal smelting, vehicle emissions, paints, dissolution of minerals, combustion of fuels and fertilizers, has released many of the toxic metal to the environment (Balasubramanian et al., 2009; WHO, 2010).

Average concentrations of elements in the earth's crust and the top soils have been reported in literature by several sources. The data are pooled from several published work worldwide. Table 1.1 summarizes some of the commonly measured elements in soil sample (Alloway 2013; Kabata - Pendias 2011).

Table 1.1: Availability of some heavy metals in earth's crust and top soil (Alloway 2013; Kabata 2011).

Element	Earth's crust availability (mg/Kg)	Top soil availability (mg/Kg)
Cd	0.1-0.5	0.07-1.1
Co	10-12	1-40
Cr	100	60
Cu	25-75	14-109
Hg	0.58-1.8	0.020-0.625
Mn	350-2000	411-550
Ni	20	13-37
Pb	10-30	27
Zn	20-200	60-89

1.2. Heavy Metals Essentiality and Toxicity

Top soil is a vital resource, which affects life of all organisms. Soil plays a significant rule to keep all environments' elements in balance with each other (Kabata - Pandias 2011). However, since the beginning of the industrial revolution, this balancing start to be distorted, due to the huge increase in pollutants accumulation, and inability of soil to accommodate this large amount of contaminants. Moreover, metals are different from organic pollutants as they don't degrade with the time. However; they can transform from form to another form. Therefore, the release of even very small levels of a certain element, will accumulate with time to reach harmful levels to the environment, and thus, transmitted to human via food chain (Anyakora, et al., 2013; Fazeli, 2009).

From a biological point of view, trace elements are classified into three groups, namely: essential, non-essential, and toxic elements (Alloway and Ayres, 1997; Monni et al., 2000; Lenntech, 2011).

Essential Elements: This group of elements is required by most living organisms in small – but critical – amounts for normal and healthy growth. However, excess concentrations of these elements causes toxicity, and also their deficiency causes disease under normal living conditions. This group of elements includes: Fe, Zn, Co, Cr, Cu, Mn and Mo (Alloway

and Ayers, 1997). Figure (1.1-a) shows the relationship between the plant yield and concentration of essential trace metals.

Non-Essential Elements: These elements are always found in the body tissues and fluids, but no proof of essentiality has been established for them. They cause toxicity at concentrations that exceed the tolerance level of the organism but do not cause deficiency disorders at low concentrations. Examples of these elements are; Li, B, Ge, Rb, and Sr, which are clearly shown by typical dose-response curves (Figure 1.1-b) (Lenntech, 2011; Monni et al., 2000).

Toxic Elements: These elements are not needed by the organisms, and they are extremely toxic even at very low concentrations, such as Cd, Hg, Pb and Au (Figure 1.1-b). It should be noted, however, that all trace elements are predominantly toxic when their level exceed the limits of safe exposure. This limit varies widely from element to another.

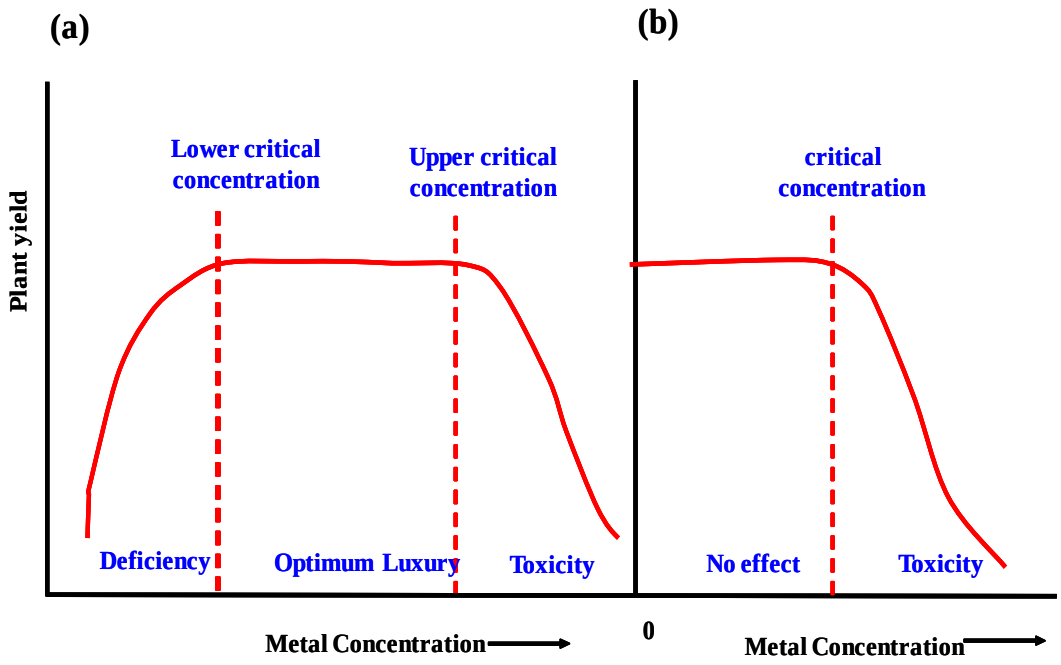


Figure 1.1: Typical dose –response curve for essential trace elements (a) and non-essential and toxic trace elements (b). (Adapted from Alloway, 1997).

1.3. Sources of Heavy Metals in the Environment

Heavy metals are naturally occurring elements that are found in the soil, rocks and earth crust. However, heavy metals that are released to the environment from natural sources are usually present in hard matrices that are difficult to be leached to the soil or aquatic systems. On the other hand, elements that are released to the environment by various human (anthropogenic) activities (such as; mining, smelting operations, industrial production, domestic and agricultural) are usually present in an easy-to-release matrices and can be easily utilized by various biological systems.

For these reasons, it is of particular important to classify heavy metals in the environment according to their sources.

Heavy metals in the environment can be attributed to two major sources (Goyer et al., 2001; He et al., 2005; Koch and Rotard, 2001; Raikwar et al., 2008; Tchounwou 2012).

Anthropogenic sources; which include:

- Fossil fuels and coal combustion
- Industrial effluents
- Solid waste disposal
- Fertilizers and agricultural use
- Mining and Metal processing
- Burning of waste materials
- Smelting operations
- Metal corrosion
- Other sources (electronics, paints, nuclear stations … etc).

Natural sources; which include:

- Weathering of mineral deposits
- Volcanic eruptions
- Windblown dusts.

1. 4. Effects of Some Heavy Metals on Human Health

Lead: Lead is one of the most toxic metal pollutants. Lead affects the central nervous system, the kidneys and blood cells. It can lead to hypertension, reproductive toxicity and developmental effects. High concentrations of lead can cause irreversible brain damage (encephalopathy), memory problems, irritability, coma and death if not treated immediately (Jarup, 2003).

Chromium: All forms of chromium can be toxic at high levels, but chromium (VI) is more toxic than chromium (III). Breathing high levels of chromium (VI) can damage the nose while ingesting it can cause stomach ulcers, convulsions, and kidney and liver damage. Chromium is classified as a human carcinogen. Long-term exposure can cause kidney and liver damage, and damage too circulatory and nerve tissue (ATSDR, 2012a).

Copper: Copper is an essential substance to human life. High levels of copper can cause anemia, liver and kidney damage, and stomach and intestinal irritation (US.EPA/IRIS, 1990).

Cadmium: The main sources of cadmium are cladding industry, where it is used in protective metal coating, nuclear reactors, alkaline cells

and alloy industry. High exposure to cadmium may cause injury to lungs, death result from pulmonary edema, hypertension, possible prostate cancer, lung cancer (from inhalation exposure), and shortness of breath. It affects liver, kidney, and pancreas (ATSDR 2012b; Martin, 2003).

Nickel: Nickel is a very abundant element. In the environment, it is found primarily combined with oxygen (oxides) or sulfur (sulfides). Small amounts of nickel are needed by the human body to produce red blood cells, however, in excessive amounts, can become mildly toxic. Long-term exposure to nickel can cause decreased body weight, rapid heart rate, liver damage, cancer, and skin irritation (Pouls, 1999; Jordan, 2007).

Zinc: Zinc is one of the most common elements in the earth's crust. Zinc is found in the air, soil, and water and is present in all foods. High levels of zinc can cause Inhalation of fumes may result in sweet taste, throat dryness, weakness, generalized aching, and vomiting (DHFS, 2000).

Cobalt: Cobalt is a compound that occurs in nature. Cobalt is released from burning coal and oil, and from car and truck exhaust. Vitamin B12 is a cobalt-containing organometallic compound that is essential for good health. Acute toxicity of cobalt may be observed as effects on the lungs, including asthma, pneumonia, and wheezing, that have

been found in workers who breathed high levels of cobalt in the air (Imray, 2001; DHFS, 2000).

1.5. Common Analytical Techniques for Trace Metals Analysis

Environmentalists are performing researches on trace and heavy metals have selected the most appropriate techniques among wide applications of analytical techniques and methods. In making this choice, several factors play a role, such as; the objectives of the study, level of sensitivity and accuracy obtainable with the analytical technique, but equally important are economical and practical considerations, including price, ease of operation, speed of analysis and total cost per analytical result obtained layout (Ebdon et al., 1998; Broekaert, 2002).

The most commonly used analytical techniques for measuring trace elements are; X-ray Fluorescence (XRF), Atomic Absorption Spectrometry (AAS), Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) and Inductively Coupled Plasma Mass Spectrometry (ICP-MS). In addition to these techniques, Ion Chromatography (IC) is commonly used as an analytical technique for the analysis of anions. Flame Atomic Absorption Spectroscopy (FAAS) technique was used in this study. A number of textbooks and many review articles on the theory and practice of

this technique are available. So, a brief review for the technique will be presented.

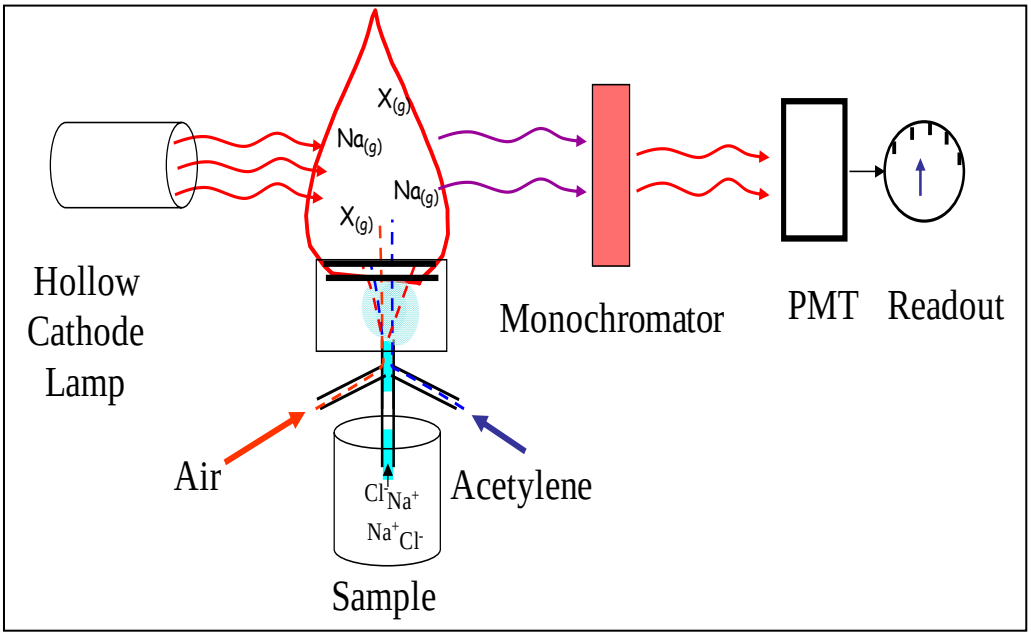


Figure 1.2: A schematic diagram for a flame atomic absorption spectrometer.

To be analyzed by FAAS, solid samples have to be digested and brought to the aqueous phase. After that, samples are aspirated by suction and nebulized into the flame at high temperature (Figure 1.2). In the flame, most atoms will be in the ground (unexcited) state. When the light from the specific hollow cathode lamp pass through the atoms in the gas phase, absorption take place and a transition from the ground state to the excited state occur (Ebdon et al., 1998; Broekaert, 2002; Christian, 2004).

1.6. Iron smelting

Melting of scrap metal is of increasing interest to the metallurgical industry since it allows the recycling of metals to clean the environment and to reduce the production cost. Obsolete scrap which consists of iron or steel products discarded after the end of their service life. Post-consumer steel products include old passenger cars, steel cans, electric appliances and other items. Obsolete scrap is often mixed or coated with other materials, such as copper, zinc, tin, glass and plastics. For this reason the content of tramp elements in obsolete scrap is usually high. Moreover, the chemical composition of obsolete scrap fluctuates widely depending on its origin and degree of processing. Obsolete scrap, especially the one originating from old passenger cars, is usually processed by shredding (Janke et al, 2000 and Ostman, 2006).

There are several processes for melting steel scrap. Today, the melting usually takes place in electrical furnaces, where the electric arc furnace is the most used. Electric current can be used for heating steel in different ways. These methods are based on resistance heating for steel melting operations (Janke, 2000). During melting dust is produced in the electric arc furnace and carried away in the off-gases which contain mainly iron, but also valuable non-ferrous metals such as zinc or lead. So not only

ecological considerations, but also economy require the collection of the off-gases and their de-dusting to recover valuable material.

The Chinese Iron Smelting Factory (CISF) is located in Mafraq. The plant is based on treatment of scrap iron collected from different areas in Jordan. The melting process in this factory is based upon the use of large electrical furnaces. This process results in the burning of iron scrap materials and all other co-exciting materials. Depending on the source of the scrap, different organic and inorganic materials may join together with the target metallic phase. These may include; paint, residual fuel in old cars and its tanks, car exhaust materials, rubbers, different kinds of polymeric materials, … etc. Burning of these various materials at high temperature leads to the emission of many organic and inorganic compounds, in addition to the carbon soot and fly ash through the chimneys into the atmosphere and then move to the soil. The factory uses water-based spray filters inside each chimney to reduce the emission of these pollutants.

1.7. Literature Background

Recently, several studies have been reported in literature dealing with the increased levels of heavy metals around various types of industrial complexes. In these studies, soil in industrial areas, waste water, plants and animals have found to pose potential human health hazard. In fact, there is

an increasing awareness that heavy metals present in soil may have negative consequences on human health and environment. Due to this effect, many studies were done to understand the nature and sources of these metals in industrial areas and urban soils (Gleyzes et al., 2002; Tessier et al., 1979).

Concentrations of Cu, Zn and Pb in the topsoil around brass industry in Jamaica have been investigated (Iyake and kakulu 2012). Using US-EPA 3050 method has been used as digested procedure. All samples were analyzed by FAAS. Concentrations of Cu, Zn and Pb were 467 ± 455, 181 ± 44 and 181 ± 201mg/Kg, respectively. Concentrations of measured elements were much lower the background levels of these elements which were; 5.4 ± 1.2, below detection limit, and 12 ± 3, mg/Kg for Cu, Pb, and Zn, respectively.

The concentration of five soil heavy metals (Pb, Co, Cr, Cu, Hg) was measured in forty sampling sites in central Transylvania, Romania (suciu et al 2008). The sampling sites are known as centers of pollution due to the chemical and metallurgical activities. The concentration of heavy metals was measured by using the Inductively Coupled Plasma- Atomic Emission Spectrometry (ICP-AES). Data were verified by using the Neutron Activation Analysis (NAA) method. Their results indicated that, the concentrations for the investigated heavy metals exceed the concentration

admitted by the Romanian guideline in some regions. Highest concentrations were found for Pb, Cu and Cr; 1521.8, 1197.6 and 1080 mg/Kg, respectively.

Heavy metals were determined in top soil around a major industrial estate co-located with a housing estate in the southwestern region of Nigerian Etim and Onianwa, (2012). Results indicated that, average levels of Cu (2850 ± 3340 mg/kg), Pb (768 ± 450 mg/kg) and Ni (105 ± 30 mg/kg) at the dumpsites were significantly higher than other locations. Generally, the degree of contamination within the vicinity of the estate was of the order Cu > Pb > Ni > Cd > Co > Cr. Average total bioavailable (non-residual) fraction of the metals consisted of 52.0% of the metal load, with this fraction for Cd and Pb being 73.9% and 60.7%, respectively.

The baseline concentration levels of Cd, Mn, Cr, Ni, Cu, Zn and Pb were determined in the top soil around the iron ore deposit at Itakpe North Central Nigeria (Stephen and Oladele 2012). One hundred and sixty soil samples were randomly collected over an area of 5x5 Km^2 between December 2003 and November 2005. The soil samples were subjected to standard methods and analyzed using FAAS. The soil metals concentration sequence was Zn > Mn > Pb > Ni > Cu > Cr > Cd, with Zn having the highest relative abundance in top soil in respect to the measured metals

while Cd had the least. There was decreasing gradient in the heavy metals concentrations from top soil (0–15cm) to depth 110cm.

Heavy metal concentrations were examined in top soil samples to evaluate a cement plant and an industrial waste incinerator as pollution sources Cordoba city (Argentina) (Bermudez et al., 2010). The following elements; As, Ba, Ca, Ce, Co, Cr, Cs, Eu, Fe, Hf, K, La, Lu, Na, Nd, Rb, Sb, Sc, Se, Sm, Ta, Tb, Th, U, Yb and Zn were measured by Neutron Activation Analysis (NAA), and Co, Cu, Fe, Ni, Pb and Zn by FAAS. Concentrations of measured elements are generally related to wind transportation from the industrial area in the north of the city. Levels of Cu, Pb and Zn were more influenced by the cement plant. Concentrations of Ca, Ba and Fe were related to the distance from the cement plant.

The contamination of soil and plants from area around a cement factory in Riyadh, Saudi Arabia, has been investigated by Al-Oud et al., (2011). Total concentrations of Cd, Cr, Cu, Pb, Zn, Fe and Al in plant and soil samples were determined by ICP-AES. Results of this study revealed that; concentration of Cd, Cr and Zn metals in the soil surface were higher than in subsurface soil samples, while Cu, Pb and Fe concentrations in subsurface soil were higher. The authors have attributed the accumulation of heavy metals on the surface soil to the soil chemical and physical properties, such as high CaCO3 % as well as alkaline PH value. The

Enrichment factor (EF) values of heavy metals in soils ranged from 0.25 to 830. Moreover, high levels of metals especially Fe and Al in unwashed samples of plants were found, due to exposure to aerosols.

Krishna et al., (2013) have presented the contamination load of toxic heavy metals (As, Ba, Co, Cr, Cu, Mo, Ni, Pb, Sr,V, Zn and Zr) in 57 soil samples collected around three different active chromites mining sites around Karnataka, India. Soil samples were analyzed for heavy metals by using wavelength dispersive X-ray fluorescence spectrometry. Results indicated that the study area is contaminated due to chromium metalliferous mining in various places. The detected levels of total metal contamination in many of samples were found to exceed international threshold values. Risk assessment based upon soil quality guidelines limits proves that the soil is a serious health risk to humans.

Speciation and spatial variations of heavy metal deposition in agricultural soils of two rural communities that are affected by a mining area near Marrakech city (Morocco) were investigated (Barkouch and Pineau, 2015). Soil contamination assessment was carried out by measurement of the total metal concentration of Cd, Cu, Pb and Zn. In this study, they found that, Pb and Zn were mostly associated with long-term mobile fraction; Cd was mostly associated with the exchangeable fraction, while Cu was widely abundant in the oxidized form. The apparent mobility

and bioavailability of these four metals in the studied soils were in the order: Cd >Zn>Pb>Cu. This means that cadmium has the highest mobility in the studied soils.

Levels of some heavy metals were investigated in soils and plants in southeast Korea, around a copper-tungsten mine (Jung, 2008) to discuss the impact of past base metal mining on the surface environment. The samples were analyzed by ICP-AES for Cd, Cu, Pb and Zn. Average concentrations of heavy metals in surface samples in the mine dump sites were: Cu, 1953 mg/Kg; Pb, 1030 mg/Kg; Zn, 419; mg/Kg; Cd, 4.4mg/Kg. The results of chemical analysis indicated that the heavy metals in soils decreased with distance from the source, controlled mainly by water movement and topography.

In Jordan, few case studies have been found in literature about the influence of the industrial plant on the surrounding areas. For example, Al-khashman and Shawabkeh, (2006), have studied metals distribution in soils around the cement factory in southern Jordan. The results indicated that; high concentration of Pb, Zn, Cd, Fe, Cu and Cr were found in areas that are close to the cement factory. Concentration of measured elements was in the following order: Fe> Pb>Zn> Cd> Cu>Cr.

Speciation and spatial variations of some heavy metals in top soils around one of the Jordanian factories have been investigated (Abu Baker,

2015). In thier study, Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn in soils around the Jordan Bromine Company have been estimated. Samples were totally digested and some of them are sequentially extracted by using the standard BCR procedure and analyzed by FAAS. Average concentrations of measured elements were; (6885.6, 162.02, 146.52, 140.37, 54.69, 22.36, 12.56, 11.92 and 6.59) mg/kg, for Fe, Pb, Zn, Mn, Co, Ni, Cu, Cr and Cd, respectively in soil samples from inside the factory. While average concentration of Fe, Pb, Zn, Mn, Co, Ni, Cu, Cr and Cd in the outside soil of the factory were; (4607.42, 149.25, 44.46, 142.11, 55.55, 23.71, 8.33, 24.39 and 6.79) mg/kg, respectively. Results of speciation study indicated that, significant fractions of Cd, Mn, Ni, Pb and Co (56%, 55%, 36%, 30% and 24%, respectively) were contained in the bio-available (exchangeable) fraction. However, most of the observed Fe, Cr, Co, Pb and Zn (88%, 62%, 60%, 51% and 36%, respectively) were tied up with silicate phase (residual). This trend was almost the same for both inside and outside soil samples.

1.8. Objectives

The objectives of the present study are:

1- To estimate the levels of selected heavy metals (Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn) in soil around one of the Jordanian factories (Chinese Iron Smelting Factory) in Mafraq governorate, Jordan.

2- To assess the bio-available fraction of measured elements in collected soil samples by using the BCR sequential extraction procedure.

3- To correlate metal levels with distance from the factory.

Chapter 2

Experimental

2.1. Sampling Site

The sampling site is located in one of the villages in Mafraq governorate, north of Jordan (Latitude 32° 23' 14.68"; longitude 36° 16' 9.17"). As mentioned in chapter one, the influence of one of the iron smelting plants on the surrounding area will be investigated. Therefore, soil samples were collected from different areas within vicinity of the plant. Soil samples were collected from all directions around the plant. A map of study area and the sampling points is given in Figure 2.1.

2.2. Samples Collection

Soil samples were collected from inside and outside the plant area during (February 2014 and August 2015). All samples were collected from the top 0-15 cm of the soil (surface layer). The total number of samples was thirty six. These include; 32 soil samples and 4 sedimentary ash samples. Collected samples were directly picked up and packed in plastic bags and labeled, then sent to the laboratory. All samples were air dried for few days and then crushed with mortar and pestle. After drying, samples were sieved through 250µm.

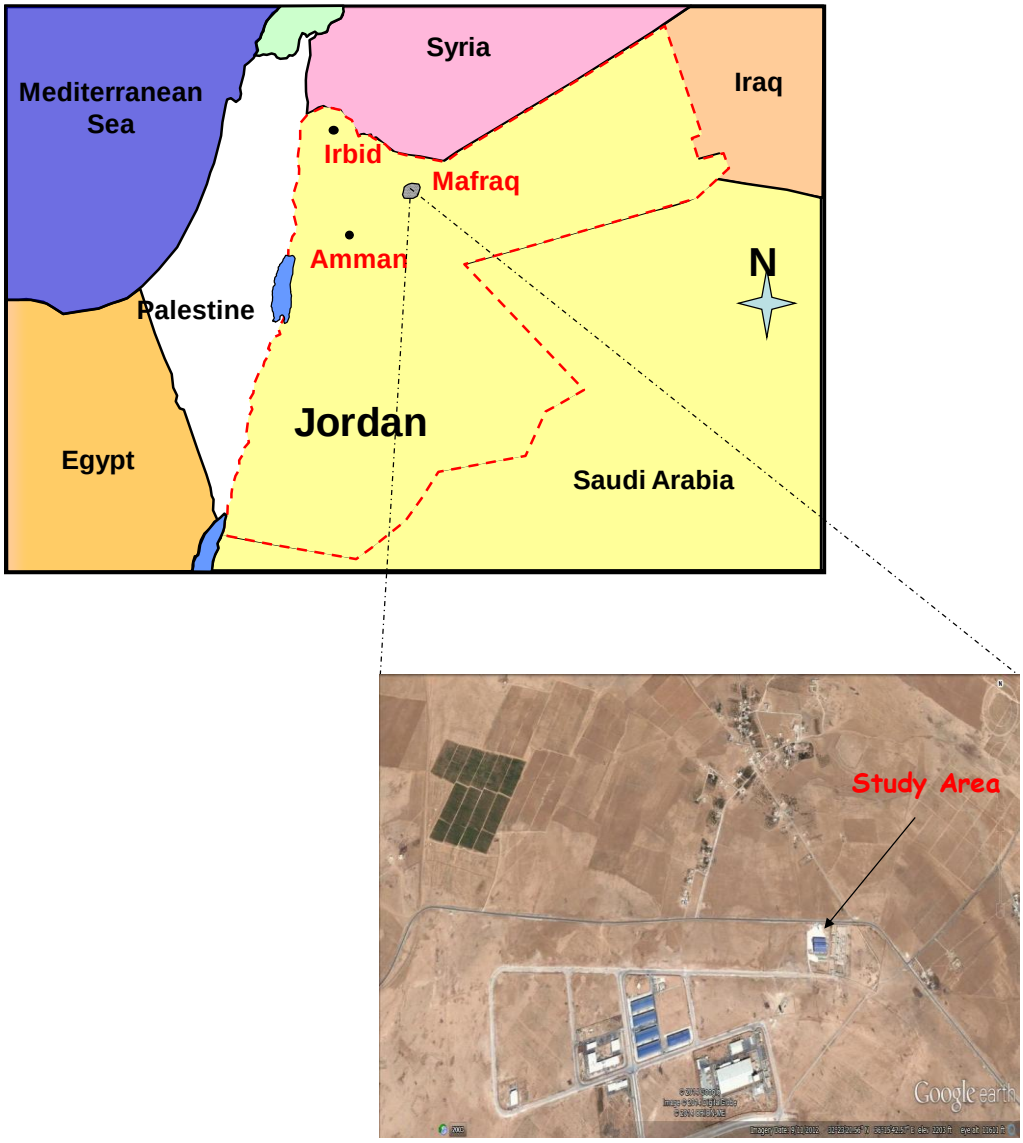


Figure 2.1: A map for the study area

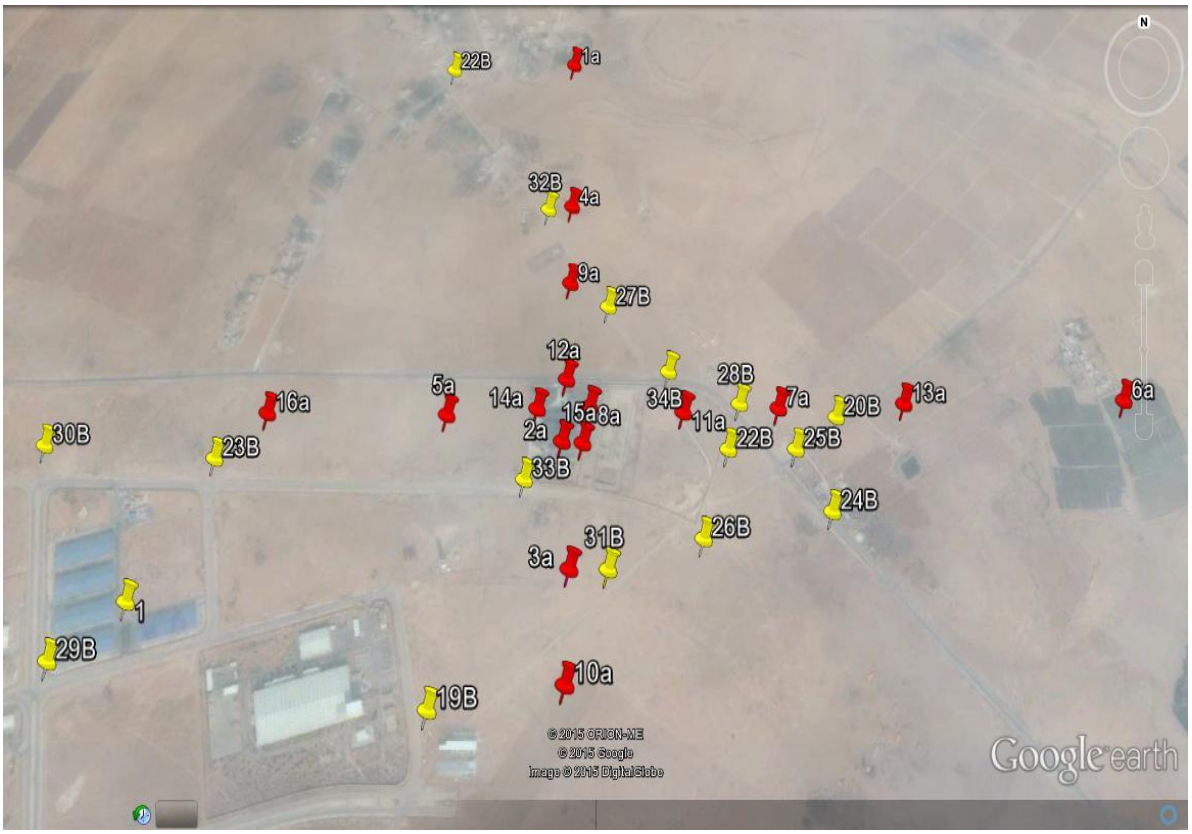


Figure 2.2: Distribution of sampling points around the iron smelting factory

2.3. Reagent and Glassware

All glassware and plastic bottles used were initially cleaned with soap, washed thoroughly with tap water, rinsed with deionized water. To remove any possible contamination by metals, the bottles used were soaked overnight in 10% HNO_3 and then washed by distilled deionized water (DDW) before drying. The chemicals and reagent used in this study are listed in table 2.1.

Table 2. 1. Chemical and reagent used in this study.

	Name	Formula	Purity
1	Acetic aced	CH_3COOH	99.5%
2	Ammonium acetate	NH_4OAc	96%
3	Hydrochloric acid	HCl	36%
4	Hydrofluoric acid	HF	48%
5	Hydrogen peroxide	H_2O_2	30%
6	Hydroxylamine hydrochloride	$NH_2OH.HCl$	99%
7	Nitric acid	HNO_3	Re-distilled, 65%

2.4. Sequential Extraction Method

Sequential extraction technique is commonly used to fractionate the solid-phase forms of metals in soil. Many sequential extraction schemes have been developed for the determination of binding forms of trace metals in soil. The lake of uniformity of these schemes, however, did not allow the results to be compared worldwide. As a result, in the 1990_s, a group of experts, working under the sponsorship of the commission of the European Communities, Community Bureau of Reference (BCR) met with the aim of harmonizing extraction methodology (Prohaska et al., 2004).

The main outcomes were the development of the standard sequential extraction procedure (BCR). The BCR partitions elements into four operationally defined geochemical fractions including acid extractable, reducible, oxidizable and residual (Gleyzes et al., 2002). The detailed procedure for each of these steps is given below:

Step 1- Exchangeable and Weak Acid Soluble Fraction

The metal fraction recovered in this step may be present as co-precipitated with carbonate minerals but also as specifically sorbed to some sites of the surface of clay.

i) A 1 g soil sample was treated with 40 ml of 0.11 mol L^{-1} acetic acid solution.

ii) This mixture was shaken in a mechanical shaker at 30± 10 rpm at 22± 5 ˚C TAB.

iii) The extract was separated from the solid residue by centrifugation at 3000 rpm for 20 min.

iv) The supernatant was decanted, collected in polyethylene bottle and stored at 4˚C until analysis.

v) The residue was washed by adding 20 ml of (DDW), shaking for 15 min on mechanical shaker, and centrifuging at 3000 rpm for 20 min.

vi) The supernatant was decanted, and discarded, taking care not to discard any of the solid residues.

Step 2- Reducible Fraction

In this fraction, hydroxylamine hydrochloride is used as reducing reagent. Hydroxyl amine hydrochloride has ability to dissolve different metallic oxides depends on pH, concentration, extracting time and temperature.

i) 40 ml of 0.5 M hydroxylamine hydrochloride solution was adjusted to pH 1.5 and added to the residue from the first step.

ii) This mixture was shaken in a mechanical shaker at 30± 10 rpm at 22± 5 ˚C TAB. The extract was separated and stored as in the first step. The residue was washed as in the previous step and the supernatant was decanted and discarded.

Step3- Oxidizable Fraction

In this fraction hydrogen peroxide (H_2O_2) is generally used. The oxidation process is promoted be heating for several hours, H_2O_2 destroy organic matter, and sulphides.

i) 10 ml of 8.8 M of H_2O_2solution was carefully added to the residue from the second step.

ii) The mixture was digested for 1 h at 22 ± 5 ˚C, and the volume of the liquid was reduced to less than 3 ml.

iii) A second aliquot of 10 ml of H_2O_2 was add, the mixture was digested for 1 h at 85± 2 ˚C, and the volume was reduced to about 1 ml.

iv) Finally, 40 ml of 1 M ammonium acetate adjusted to pH 2.0 ± 0.1 was added to the cool, moist residue.

v) This mixture was shaken in a mechanical shaker at 30± 10 rpm at 22± 5 ˚C. The extract was separated and stored as in the previous steps. The residue was washed as in the previous steps and the supernatant was decanted and discarded.

Step 4- Residual Fraction (Modified)

Primary and secondary minerals containing metals in the crystalline lattice constitute the bulk of this fraction. Its destruction was accomplished achieved by digestion with strong acids. In this step the residue from step 3 was digested with (HNO_3 + HCl + HF + H_2O_2) following the same procedure that have been developed and used for total digestion (next section).

i) 9 ml of high purity concentrated HNO_3 and 2 ml of HCl were added to the residue from the third step then were transferred to inert polymeric microwave vessel.

ii) After that, 6 ml of HF was added to dissolve siliceous materials.

iii) Then 1.0 ml of H_2O_2 was added to the mixture, and then the vessels were put for 25 minutes using microwave heating with (Multiwave Eco-Anton Paar) microwave system.

iv) The digest was diluted with DDW up to 50 ml and stored in plastic bottles.

2.5. Total digestion

The procedure described by Kingston, et al, 1988- 1997 (EPA 3052) was followed to digest the samples with some modifications. In this procedure, 9 ml of high purity concentrated HNO_3, 2 ml of HCl, 3 ml of HF and 1 ml of H_2O_2 were added to 0.5 g of the sample, respectively. The vessels were then heated using microwave (Multiwave Eco-Anton Paar) to 180˚ C over 5.5 minutes and then held at 180˚ C for at least 9.5 minutes (Kingston and Walter, 1997). This procedure was applied to a number of different samples and tow of standard reference materials (SRM) samples.

All tested samples were not completely dissolved. Therefore, different ratios of the reagents used above were tested using different microwave temperature programs to ensure complete digestion of the samples. During these steps, it was clear that the siliceous materials in the samples are high, therefore, larger amount of HF and longer contact time are needed to damage the hard siliceous matrix. Finally, complete digestion was achieved using the following ratio and program; 9 ml of high purity concentrated HNO_3, 2 ml of HCl, 6 ml of HF and 1 ml of H_2O_2 were added to each vessel. The temperature profile is specified to reach 180 ± 5˚C in approximately less than 5.5 minutes and remaining at 180 ± 5˚C for 20 minutes. The digest was cooled and diluted with DDW up to 50 ml and stored in plastic bottle.

2.6. Analysis of Samples

Concentrations of Cd, Co, Cr, Cu, Fe, Mn, Ni and Zn were determined in all samples by using a flame atomic absorption spectrophotometer FAAS (Varian Spectra AA, AUSTRLAIA). Measurements were made using specific hollow cathode lamp for each element. The proper wavelength and the slit width were adjusted and other FAAS conditions employed in these determinations are summarized in Table 2.2. The flame used for all elements was air-acetylene.

A stock solution, 1000 ppm, (BDH chemicals Ltd, Phoole, England) was used to prepare the working standard solutions for each element. These standards were tested by comparing them to commercial multi-element standard and were always in excellent agreement. Calibration curves are shown in Figure 2.2.

Table 2.2: The basic instrumental parameters for FAAS					
Element	λ (nm)	Air flow ($L.min^{-1}$)	Acetylene flow($L.min^{-1}$)	Slit width (nm)	Lamp current (mA)
Cd	228.2	3.5	1.5	0.5	4.0
Co	240.7	3.5	1.5	0.2	7.0
Cr	357.9	3.5	2.5	0.2	7.0
Cu	324.8	3.5	1.5	0.5	4.0
Fe	392.0	3.5	2.0	0.2	5.0
Mn	279.5	3.5	2.5	0.2	5.0
Ni	232.0	3.5	1.5	0.2	4.0
Pb	217.0	3.5	1.5	1.0	10.0
Zn	213.9	3.5	1.5	1.0	5.0

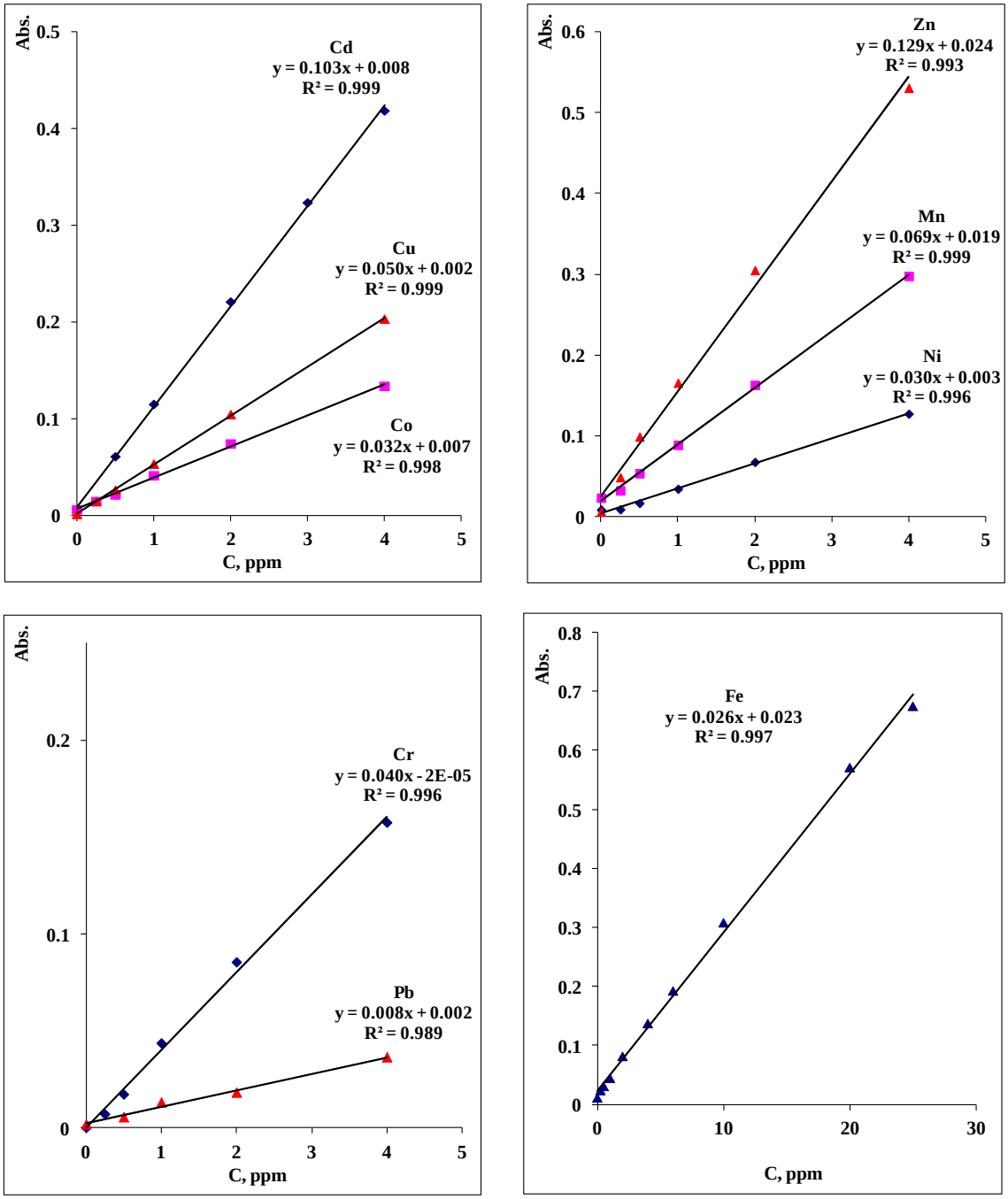


Figure 2.3: Calibration curves for measured elements, showing the Sensitivity trend of the element is; Zn> Cd> Mn> Cu> Cr> Co> Fe> Ni> Pb.

2.7. Quality Control and Quality Assurance

In analysis of real samples, particularly in samples collected for metal analysis, blank measurements play a crucial important role. Contamination by metals is a strong possibility; due to their wide spread distribution. Frequently analysis of blank is the only way to detect and correct any contamination problem.

Blank samples were prepared for each of the extraction steps and during samples digestion. Finally, blank samples were analyzed along with the samples digestion. Finally, blank samples were analyzed along with the samples and the average contribution of the added acids and reagents were estimated. All results were blank corrected by subtracting the average blank concentration observed for a certain element from the concentration observed for the same element in each sample.

The sample-to-blank ratios for measured elements were larger than 10. These findings indicate that blank subtraction does not have a significant contribution on the observed concentrations.

The detection limits (DL) of the methods were estimated as the concentration that produced three times the standard deviation at the respective wavelengths selected for each element during calibration runs. The blank samples were used to calculate the detection limits. The calculated values were as follow; 0.014, 0.028, 0.008, 0.028, 0.43, 0.006,

0.061, 0.042, 0.027 $mg.L^{-1}$ for Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb, and Zn, respectively.

The accuracy and precision of the analysis results were checked by periodic analysis of Standard Reference Material (SRM) obtained from the National Institute of Standard and Technology (NIST). Two SRMs were analyzed after being digested using the procedure as samples. Each SRM was repeated for three times. The SRM used in this study are; SRM-1646a (Estuarine Sediment) and SRM-2702 (Inorganics in marine sediment). Approximately 0.5 g of SRM was transferred into a Teflon beaker and digested using the same microwave digestion procedure and analyzed along with samples. The results for SRMs analysis were in good agreement with the certified values (Table 2.3).

Table 2.3: Accuracy and precision based on two SRMs for FAAS analysis

	Unit	SRM-1664a		SRM-2702	
		Found ± SD	Certified± SD	Found ± SD	Certified± SD
Cd	mg/Kg	0.63± 0.28	0.15 ± 0.01	0.84 ± 0.22	0.82 ± 0.01
Co	mg/Kg	5.26± 0.71	5	25.7 ± 2.16	27.76 ± 0.58
Cr	mg/Kg	39.6 ± 1.5	40.9 ± 1.9	322.2 ± 5.7	352 ± 4.7
Cu	mg/Kg	9.9 ± 1.1	10.01 ± 0.34	94.73 ± 1.63	
Fe	mg/Kg	1.95 ± 0.19	2.00 ± 0.04	7.3 ± 0.48	7.91 ± 0.24
Mn	mg/Kg	232.0 ± 2.9	234.5 ± 2.8	1405 ± 64	1757 ± 58
Ni	mg/Kg	21.5 ± 2.3	23	76.5 ± 2.8	75.4 ± 1.5
Pb	mg/Kg	10.32 ± 4.6	11.7 ± 1.2	134.6 ± 2.7	132.8 ± 1.1
Zn	mg/Kg	47.23 ± 2.9	48.9 ± 3.6	417.0 ± 11.4	485.3 ± 4.2

Chapter 3

Results and Discussion

3.1. Statistical Review

As previously mentioned in the experimental part, different types of samples (sedimentary ash and soil), were collected from the vicinity of Chinese Iron Smelting Factory in Mafraq, northern Jordan. Collected samples were sequentially extracted and totally digested to bring the target ions into the aqueous phase before the analysis by FAAS. All samples were analyzed for the following elements; Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn.

Statistical summaries for all measured elements in the inside and outside soil samples are presented in Tables 3.1 and 3.2. In addition, statistical summaries for sedimentary ash samples are also presented in Table 3.3. In these tables, average concentrations, standard deviations, geometric means, minimum and maximum concentrations of samples analyzed are presented.

Standard deviations listed in Tables 3.1, 3.2 and 3.3 are high for most of the measured elements. Such high standard deviations are expected for environmental data and do not mean poor analysis or sampling. These observed variations are due to the large variability of the locations,

variation in physical and chemical nature of the sampling point. Geometric means are included because concentrations of metals in environmental samples are more often log-normally distributed rather than normally distributed. Data sets which show lognormal distribution patterns are characterized by median and/or geometric means rather than arithmetic means (average). Therefore, both geometric and arithmetic means are added to the tables.

The average concentration of Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn in the inside factory soil samples were; 2.72 ± 3.72, 98.8 ± 19.0, 45.6 ± 59.6, 35.3 ± 11.0, 19073 ± 4301, 867 ± 240, 87.9 ± 22.6, 500 ± 251, 514 ± 407 mg/Kg, respectively. The order of the concentrations were; Fe > Mn > Zn > Pb > Co > Ni > Cr > Cu > Cd (Table 3.2). Average concentrations of Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn in the outside soil samples were; 2.58 ± 4.03, 78.26 ± 32.03, 47.54 ±33.18, 31.88 ±4.24, 14516 ± 5539, 1142 ± 398.46, 88.59 ± 27.67, 271.43 ± 172.33, 193.82 ± 75.03 mg/Kg, respectively. The order of concentrations were as; Fe > Mn > Pb> Zn > Ni > Co > Cr > Cu > Cd (Table 3.3).

As shown in Tables 3.1 and 3.2, the levels of measured elements in soil samples collected from the inside the factory area are generally higher than those collected from the outside area. The increased levels of measured elements in the inside soil samples is obviously due to the close

proximity of the sampling points to the chimneys. In all cases, Fe was the dominant element among the measured ones followed by Mn.

As expected, highest levels were observed in the sedimentary ash samples. concentration Levels of measured elements in the ash samples were as follow; 309.74 ± 155.60, 111.1 ± 32.4, 109.2 ±67.7, 382.1 ± 161.3, 22220 ± 2079, 2986 ± 958, 153.7 ± 85.1, 7509 ± 1540 and 2955 ± 132.2 mg/Kg for Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb and Zn, respectively. The concentration order of measured elements in the ash samples were as follow; Fe > Pb > Mn> Zn > Cu >Cd> Ni > Co> Cr.

The average total metal concentrations in all samples are shown in Figures 3.1-3.3. In these Figures, concentrations of Fe were divided by 100 in order to facilitate the examination and comparisons of other elements. As shown in all samples in the three different sub-groups, Fe was the dominant metal followed by Mn in the inside and outside soil samples. However, the second element was Pb followed by Zn in the ash samples. This might be due to the volatility of these two elements which are usually creating an environmental problem in scrap recycling industry (Janke et al, 2000 and Ostman, 2006).

In addition, these Figures show clearly that the emission profiles of measured elements in the inside soil samples are similar to those of the outside samples. This can be viewed clearly from the superposition of the

two figures. This is strong evidence that the outside area of the industrial complex is affected by the plant emissions.

Table 3-1: statistical summary for total concentrations of measured elements in the inside factory soil samples (mg/Kg).

Element	AVG	MAX	MIN	SD	RSD	G. Mean
Cd	2.72	9.19	0.15	3.72	136.8	0.79
Co	98.8	119.5	64.0	19.0	19.2	97.1
Cr	45.6	142.9	0.1	59.6	131.0	7.1
Cu	35.3	50.1	21.5	11.0	31.3	33.7
Fe	19073	21871	10464	4301	23	18532
Mn	867	1163	535	240	28	838
Ni	87.9	118.3	56.9	22.6	25.7	85.3
Pb	500	944	186	251	50	448
Zn	514	1258	169	407	79	405

Table 3-2: statistical summary for total concentrations of measured elements in outside soil samples (mg/Kg).

Element	AVG	MAX	MIN	SD	RSD	G. Mean
Cd	2.58	16.90	0.03	4.03	156.23	0.33
Co	78.3	126.7	29.7	32.0	40.9	71.7
Cr	47.5	127.3	0.2	33.2	69.8	29.7
Cu	31.9	38.3	21.6	4.2	13.3	31.6
Fe	14516	23376	6328	5539	38	13522
Mn	1142	2232	216	398	35	1056
Ni	88.6	132.4	20.2	27.7	31.2	83.3
Pb	271	621	83	172	64	221
Zn	194	385	105	75	39	181

Table 3-3: Statistical summary for total concentrations of measured elements in ash samples (mg/Kg).

Element	AVG	MAX	MIN	SD	RSD	G.Mean
Cd	309.74	463.86	165.64	155.60	50.24	278.97
Co	111.1	157.3	87.5	32.4	29.2	107.9
Cr	109.2	205.7	48.4	67.7	62.0	95.3
Cu	382.1	482.3	196.0	161.3	42.2	353.6
Fe	22220	24789	19875	2079	9.0	22147
Mn	2986	4272	1954	958	32	2874
Ni	153.7	268.6	63.0	85.1	55.4	135.7
Pb	7509	9590	5926	1540	20.5	7395
Zn	2955	3057	2769	132	4.5	2953

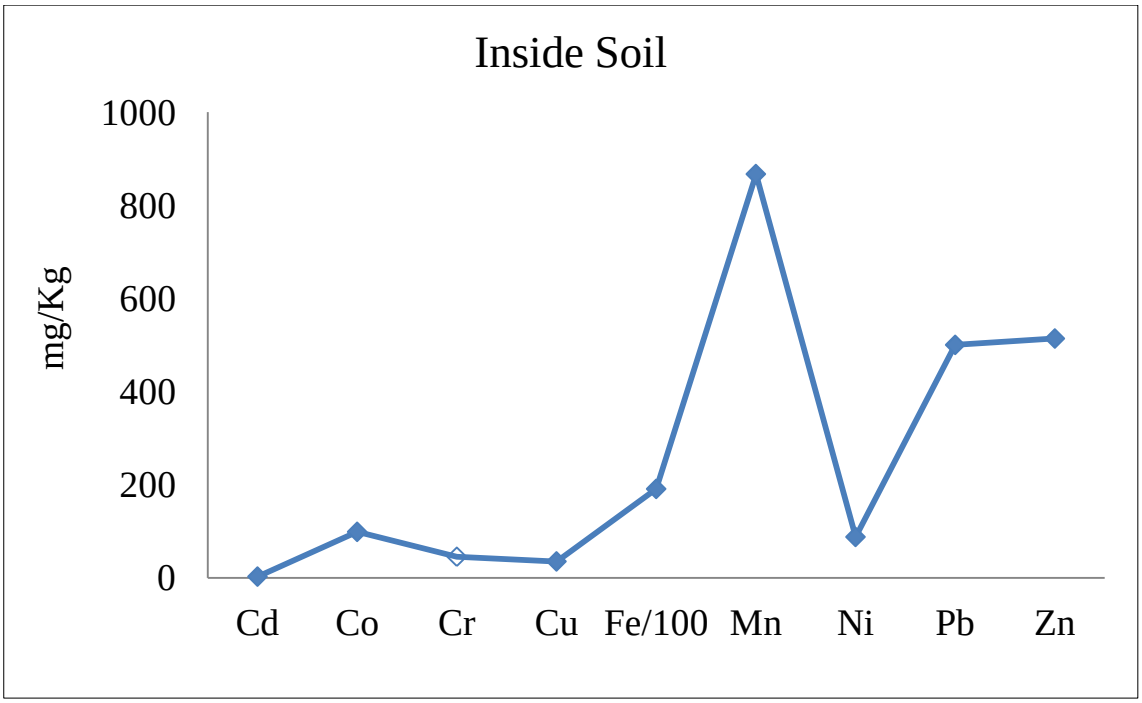


Figure 3.1. Average total elemental concentrations of the measured elements in the inside soil samples.

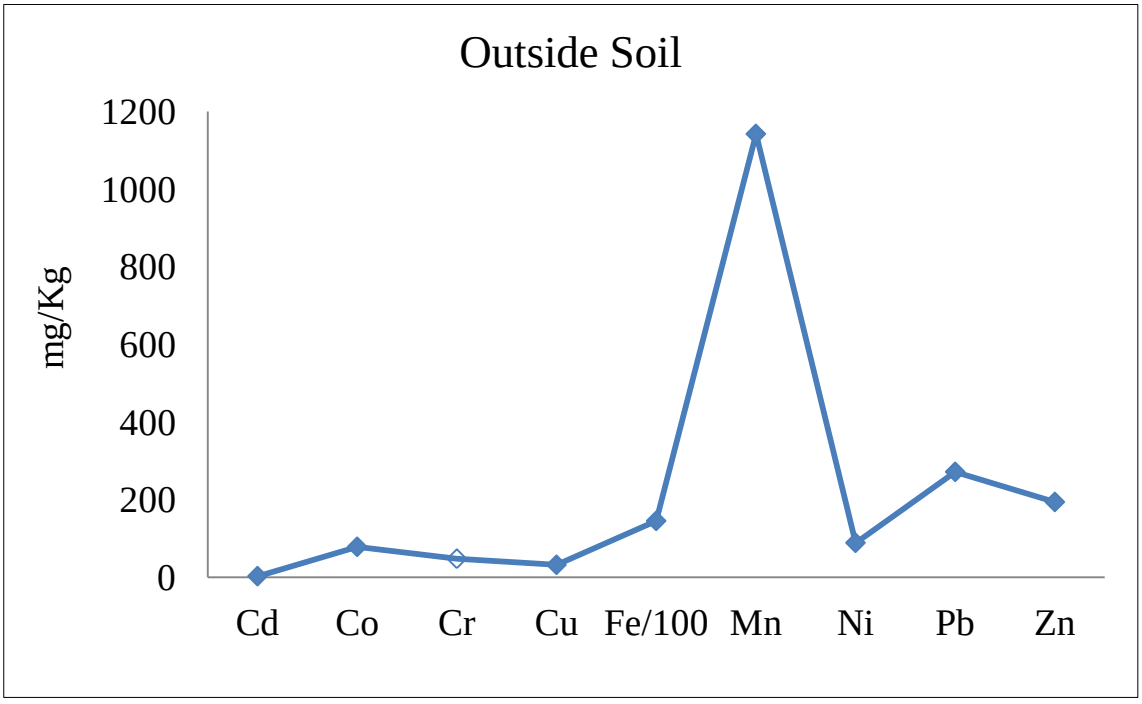


Figure 3.2. Average total elemental concentrations of the measured elements in the outside soil samples.

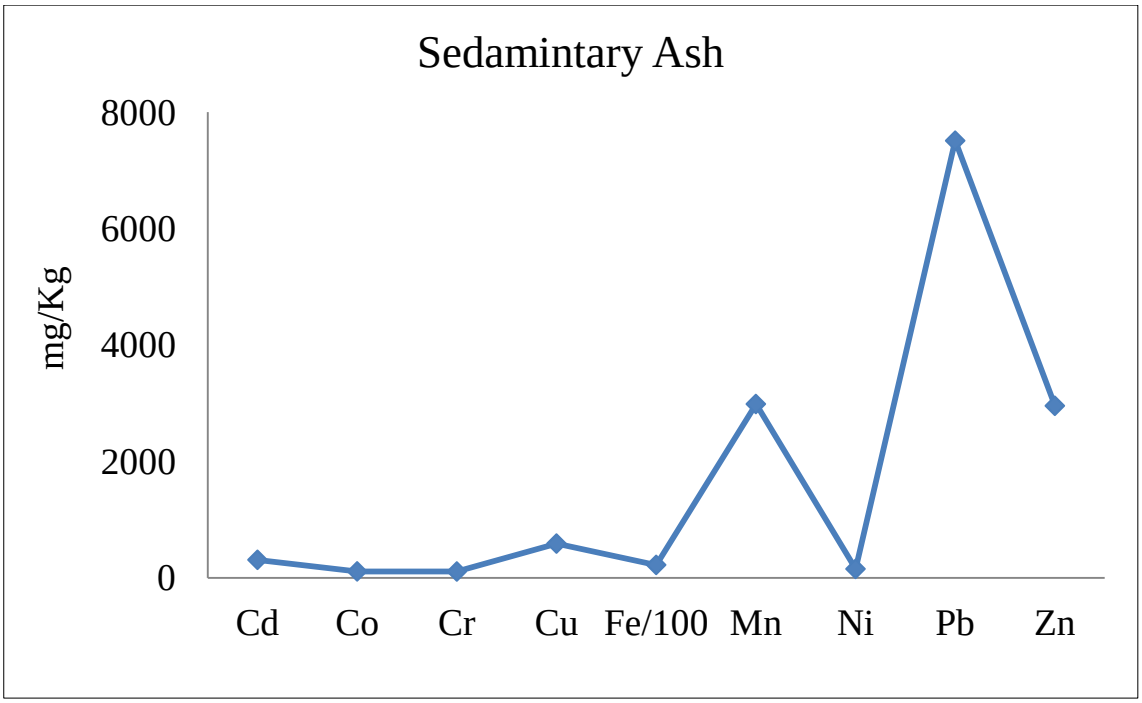


Figure 3.3.Average total elemental concentrations of the measured elements in the sedimentary ash samples.

3.2. Comparisons of Total Elemental Concentrations

Analysis of variance (ANOVA) was used to compare the mean concentration of each element in soil and ash samples. ANOVA is a statistical technique which can be used to compare means of two or more samples and estimate the different causes of variation (Kepple and Wickens,2004). A significant difference between any pair of means of samples population is concluded when the probability has a difference of 0.05 or less.

Comparisons of the total elemental concentrations summarized in Tables 3.1-3.3, indicate that the inside soil samples contain higher concentration s of Cd, Co, Cu, Fe, Pb and Zn than outside soil samples. Statistically significant differences between the levels of Fe and Mn were observed at 95% confidence levels. However, statistically significant differences between levels of Co, Zn and Pb in the inside soil samples as compared to outside at about 90% confidence level.

When comparing the inside and the outside soil samples with the contents in the ash collected close to the chimneys, a significant differences between levels of Zn and Pb was observed even at 99% CL. The same was also true for Fe, Mn and Co. As mentioned before, during melting, dust is produced in the furnace and carried away in the off-gases which contain mainly iron, but also valuable non-ferrous metals such as zinc or lead

(Janke et al, 2000 and Ostman, 2006). This may explain also the relatively high levels of these two metals in all soil samples.

Table 3.4: Calculated p-values (probability) for Student's t Test of different data sets.

Element	**Sample Type**		
	Out Vs Inside	Out Vs Ash	Inside Vs Ash
Cd	0.937	**0.029**	**0.029**
Co	**0.060**	0.129	0.529
Cr	0.940	0.164	0.178
Cu	0.488	0.083	0.084
Fe	**0.054**	**0.000**	0.164
Mn	**0.048**	**0.029**	**0.019**
Ni	0.947	0.224	0.220
Pb	0.077	**0.003**	**0.003**
Zn	0.112	**0.000**	**0.000**

- If $\mathbf{T_{cal} < T_{tab}}$ (or Pcal **> α**) then there is no significant difference in the mean of the two data sets.
- If $\mathbf{P_{cal} < \alpha}$ then there is a significant difference.
- For 95% confidence level, **α = 0.05**
- For 99% confidence level, **α = 0.01**

3.3. Speciation and Bioavailability of Metals

It is well documented that, ionic form of metal is the most toxic form of the metal. Bioavailable fraction, defined as the fraction that is available or can be available for absorption by plants or organisms from their surrounding (Peijnenburg et al., 2003).

Metals and other chemical contaminants adsorbed to soil particles have ability to accumulate in organisms. This accumulation depends on the chemical forms that the metals can be found, i.e. as simple or complex ions, as organically bound or co-precipitated with metal oxides or carbonates. Sequential extraction techniques are used to determine amounts of an element in the different phases of the soil particle by using different extraction solvents. For instant, weak acids or dilute acids are used to release metals associated with carbonate phase. This fraction is usually considered as bio-available or easily released fraction. In this study the BCR scheme, which has been described before, was used to determine the distribution of elements in the different phases; Exchangeable (carbonate), oxidizable (bounded to Fe and Mn compounds), reducible (bounded to organic and/or sulfides), residual.

The percentages of measured elements in the various phases of the soil particles are graphically presented in Figures 3.4 – 3.12 and tabulated in Tables 3.5 and 3. 6 two sets of samples are presented below; the first one

represent the soil samples collected from the close vicinity of the factory (inside) and the second set of samples represent the soil samples collected from the outside are around the factory (outside).

In these figures, fractions are labeled as; A, B, C and D which refer to Exchangeable fraction, oxidizable fraction, reducible fraction and Residual fraction, respectively. Associations of elements in different phases show large variability depending on the origin of the soil samples.

3.3.1. Availability of Cd

The Cd^{2+} ion is generally weakly adsorbed to organic matter and silicates phase. However, at pH values above 7, Cd^{2+} can co-precipitate with $CaCO_3$ or precipitate as pure $CdCO_3$, and Cd phosphates may limit the solubility as well. As a result, mobility and bioavailability of Cd in neutral to alkaline soils are low (McBride, 1994).

The largest fraction of Cd was found in the exchangeable and carbonate phase. The percentages of Cd in the carbonate phase were 51.5% and 47.9% for the outside and inside soil samples, respectively (Figure 3.4). The appearance of such high fractions of Cd in the exchangeable fraction around industrial complexes is in consistent with others in the literature (Kashem et al., 2007; El-Kammar et al., 2009; Stone and Marslek, 1996).

The next phase in which the Cd is present is the oxides phase, in which about 40% to 48% are present. Together, the exchangeable/carbonate and the oxides phases account for more than 90% of the measured Cd in both sets of samples. The appearance of Cd in these two phases is strong evidence that the major source of Cd is anthropogenic sources. Since the area is almost free of any other industrial and human activities, the iron smelter can be considered as the major source for the observed levels of Cd. It is documented that, Cd is a constituent of motor alloys or with the electronic components and batteries of cars. Because of cadmium impurities in zinc materials, cadmium also accompanies zinc oxides used in tire manufacture (Monaci et al., 2000). Therefore, the release of Cd during the melting process is highly favorable.

3.3.2. Availability of Co

The fraction distribution of Co for the inside and outside soil samples was as follow: residual> oxidizable> exchangeable> reducible. The major fraction was found was residual phase, with 43.6% and 43.8% for inside and outside samples, respectively. The oxidizable fraction was 32.1% and 29.3% for inside and outside samples, respectively (Figure 3.5, Tables 3.5and 4.6).

3.3.3. Availability of Cr

Extremely high fractions of Cr are tied up with the residual fraction. The results (Figure 3.6 and Tables 3.5 and 3.6) show that 93.4% and 83.2% of the measured Cr and present in the residual fraction for inside and outside soil samples, respectively. The partitioning of Cr followed the order: residual (silicate) > organic/sulfides phase > exchangeable ≈ Fe-Mn oxides. These results are in agreement with those reported by Durand et al., (2004), Stone and Marslek (1996) and Belzunce-Segarra et al., (1997).

3.3.4. Availability of Cu

Copper in soil commonly occurs as Cu^{2+}. Under most conditions found as adsorbed (Soil-Cu) form. which is usually more stable than other Cu minerals. Most colloidal material of soil (oxides of Mn, Fe, Al, silicate and humic materials) adsorbed Cu^{2+}strongly (Alloway, 1995).

The speciation patterns obtained for Cu in the different soil samples (inside and outside soils) are shown in Figure 3.7. As it can be seen in the Figure below, most of the copper (62% for the inside and 75.5% for the outside samples) are associated with the residual fraction. Copper is usually associated with the organic phase in literature. However, in the absence of organic and humic substances, copper is frequently associated with the silicate matrix (Alloway, 1995).

3.3.5. Availability of Fe

In both inside and outside soil samples, Fe is associated predominantly with the residual fraction. Its content in the residual fraction averaged at about 81-88% (Figure 3.8). The order of Fe percentages in different phases were as follow: residual > oxidizable > reducible > exchangeable (Tables 3.5 and 3.6). The exchangeable fraction in both sub-groups of samples was extremely low, which was below 1% in both data sets.

The occurrence of Fe in the residual fraction was abnormal. We were expecting to see most of the Fe associated with oxides phase. However, the oxides phase was the second in its content of Fe.

3.3.6. Availability of Mn

As expected, Mn is mainly associated with the Fe-Mn oxides phase. The distribution pattern of Mn in different fractions was in the order: Fe–Mn oxide > residual > organic > carbonate. Larner et al., (2006) showed that the most Mn in soil samples was associated in the carbonate phase. The same order was observed for all samples from outside soils, in which, the oxides phase is the predominant one (Figure 3.9).

3.3.7. Availability of Ni

As shown in Figure 3.10, the distribution patterns of Ni in the various phases follow the order; residual > oxidizable > reducible > exchangeable in both outside and inside soil samples. The percentage of Ni in the residual fraction averaged at about 42% (Tables 3.5 and 3.6). High percentage of Ni in residual fraction has also been reported by Adamo et al., (1996), Ma and Rao (1997) in soils, Belzunce-Segarra et al., (1997) in sediments, Stone and Marslek (1996) and Sutherland et al., (2000) in street dust. The percentages of Ni associated with carbonate phase were 8.3% and 11.7% for the inside and outside soil samples, respectively. The oxides phase accounts for about 30.5% of the observed Ni.

3.3.8. Availability of Pb

Lead exists principally as Pb^{2+} in solid soil material and soil solution. Lead is almost insoluble in reduced soils. Under oxidizing conditions Pb^{2+} becomes less soluble as pH increases. Complexation with organic matter, chemisorptions on oxides and silicate clays, and precipitation as carbonate, hydroxide or phosphate are all favored at high pH. In alkaline soils, the solubility may increase due to formation of soluble Pb-organic and Pb-hydroxy complexes. Lead is the least mobile trace metal in soils.

Relatively high fractions of Pb are associated with the exchangeable/ carbonate phase. The percentage of exchangeable phase were 31.4% for inside samples and 33.6% for outside samples (Tables 3.5 and 3.6). The distribution pattern for Pb in each fraction followed the order: exchangeable/ carbonate > residual ~ oxides > organic/sulfides phase for the inside and outside soil samples (Figure 3.11). These findings are in broad agreement with the fractionation of Pb in sediments (Pardo et al., 1990; Belzunce- Segarra et al., 1997), soils (Levy et al., 1992; Ramos et al., 1994) and street dust (Fergusson and Ryan, 1984; Stone and Marslek, 1996).

3.3.9. Availability of Zn

In this study, the distribution pattern in each fraction followed the order: residual silicate > Fe-Mn oxides > organic/sulfides phase ~ exchangeable. Majority of Zn was associated with silicate matrix, which is the most hard to release fraction (> 52%) in both sub-groups of samples (inside and outside soils) followed by the oxides phase (~25%) (Figure 3.12). Exchangeable (carbonate) fraction accounted for about 10% of the observed Zn.

Table3.5. Percent contribution of various geochemical fractions to total metal concentration in the inside soil samples.					
Element	A % Exchangeable + carbonate	B% Fe-Mn oxides	C% Organic + Sulfides	D% Residual	Total%
Cd	47.9	47.8	3.8	0.4	100
Co	14.8	32.1	9.5	43.6	100
Cr	1.6	0.3	4.7	93.4	100
Cu	7.7	24.3	6	62	100
Fe	0.2	11	7.4	81.4	100
Mn	4.6	64.3	9.1	22.0	100
Ni	8.3	30.5	19	42.2	100
Pb	31.4	26.3	13.3	29	100
Zn	9	26.3	12.3	52.4	100

Table3.6. Percent contribution of various geochemical fractions to total metal concentration in the outside soil samples.					
Element	A % Exchangeable + carbonate	B% Fe-Mn oxides	C% Organic + Sulfides	D% Residual	Total%
Cd	51.5	39.3	6.8	2.4	100
Co	17.6	29.3	9.3	43.8	100
Cr	1.5	4.7	10.6	83.2	100
Cu	7.3	11.3	5.9	75.5	100
Fe	0.2	8.9	2.4	88.5	100
Mn	6.8	59.8	10.9	22.5	100
Ni	11.7	28.1	18.7	41.5	100
Pb	33.6	22.4	17.4	26.6	100
Zn	11.7	24.5	8.5	55.3	100

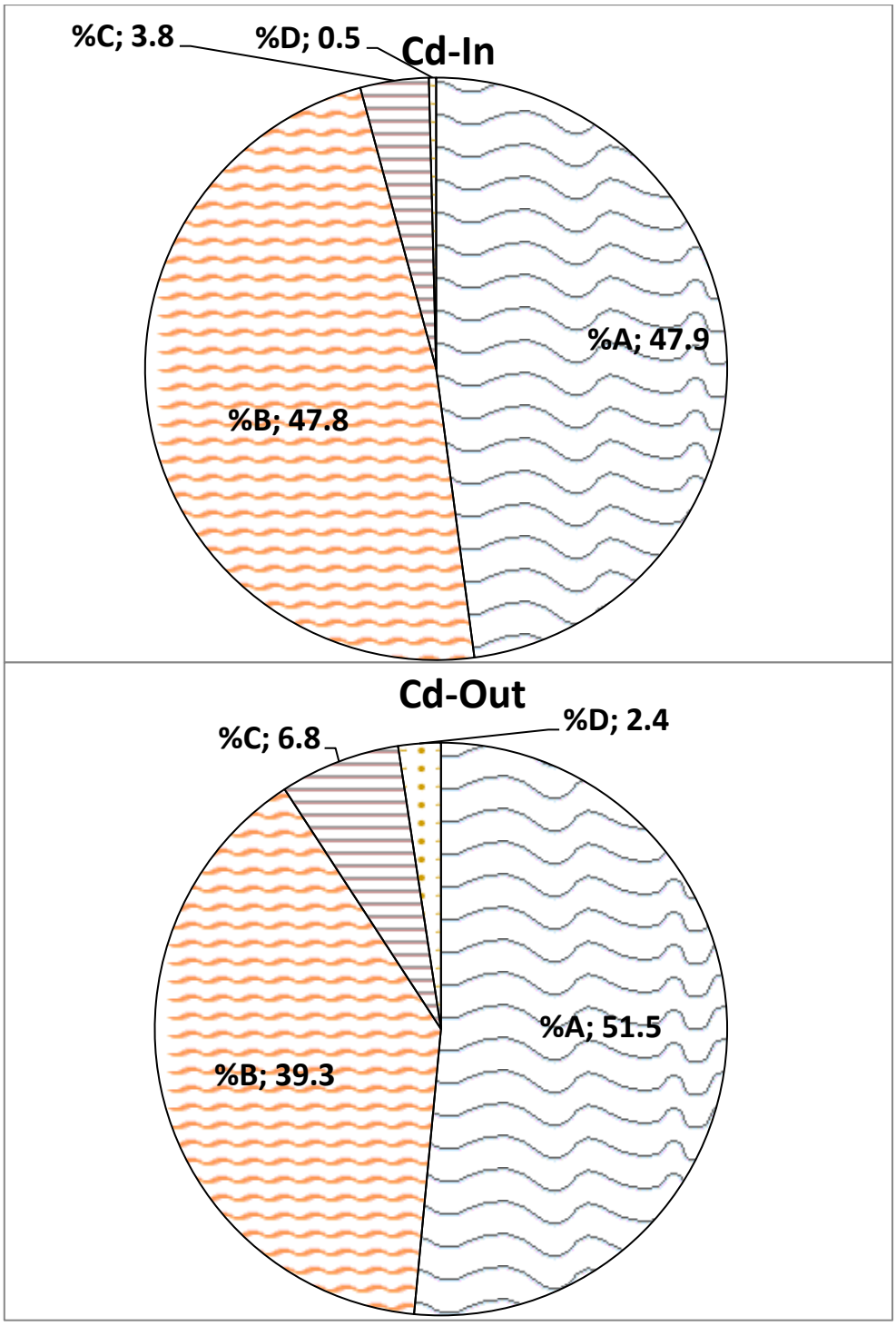


Figure 3-4: Percent distribution of Cd in various geochemical fractions to total metal concentration in inside and outside soil samples.

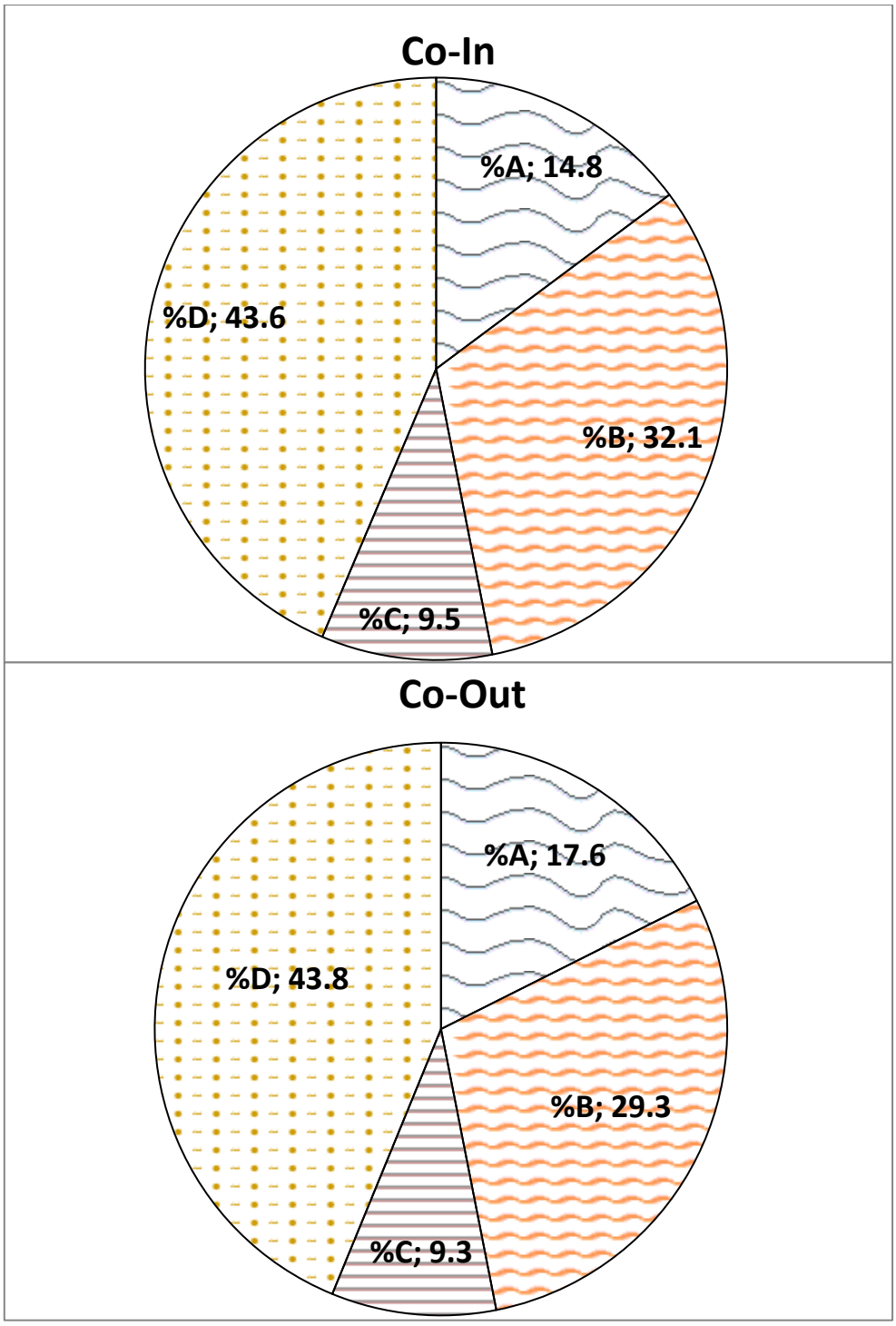


Figure 3-5:Percent distribution of Co in various geochemical fractions to total metal concentration in inside and outside soil samples.

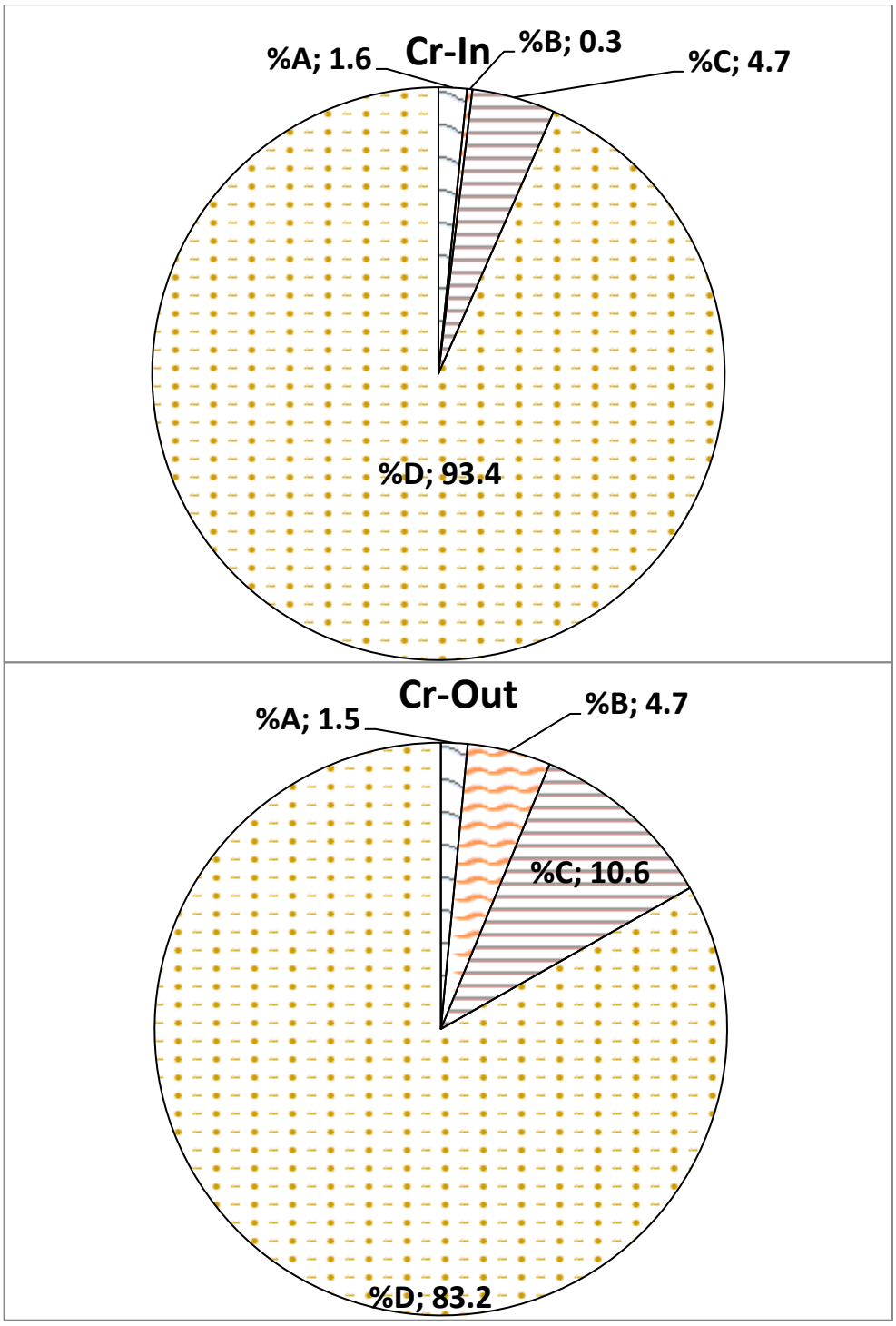


Figure 3-6: Percent distribution of Cr in various geochemical fractions to total metal concentration in inside and outside soil samples.

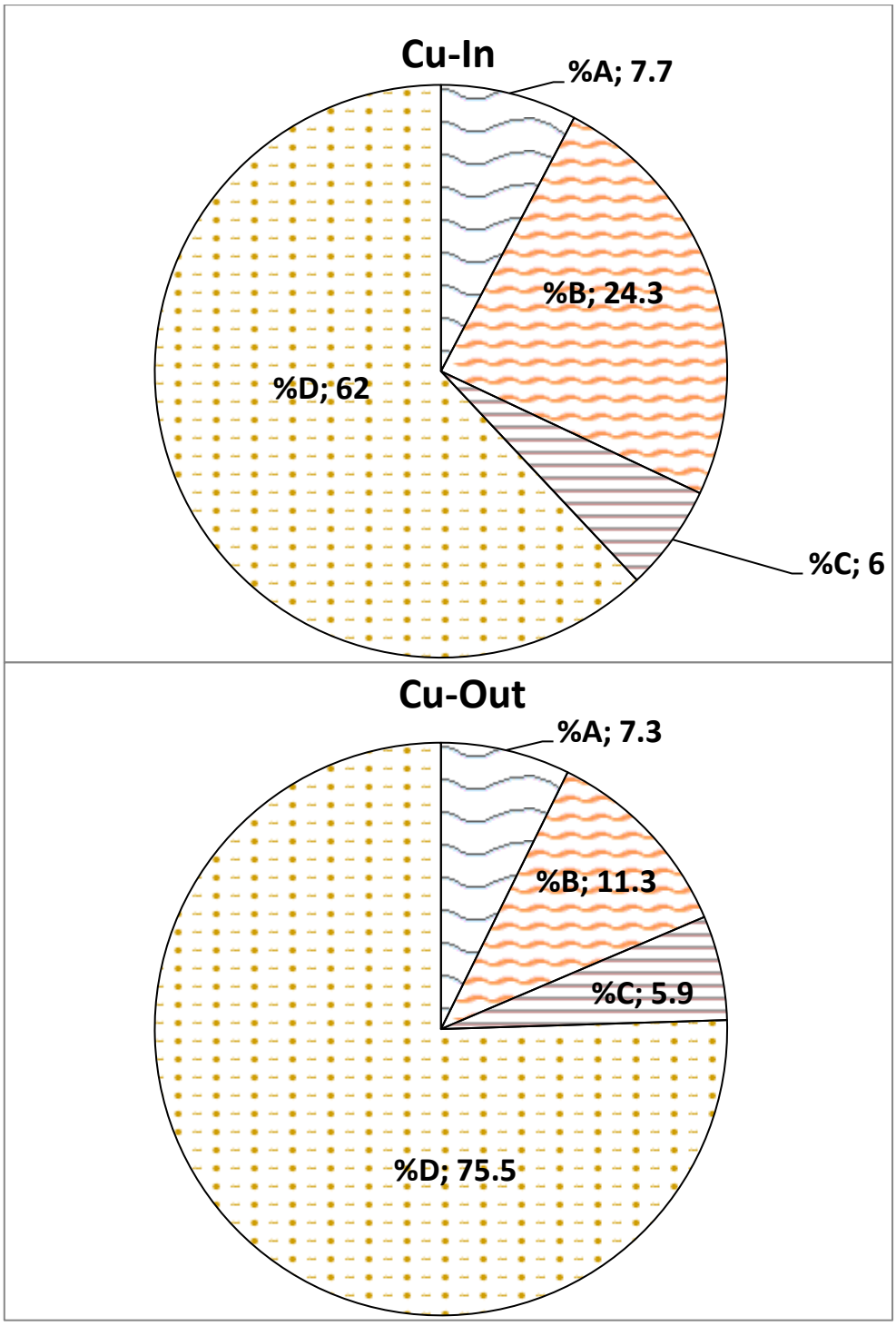


Figure 3-7: Percent distribution of Cu in various geochemical fractions to total metal concentration in inside and outside soil samples.

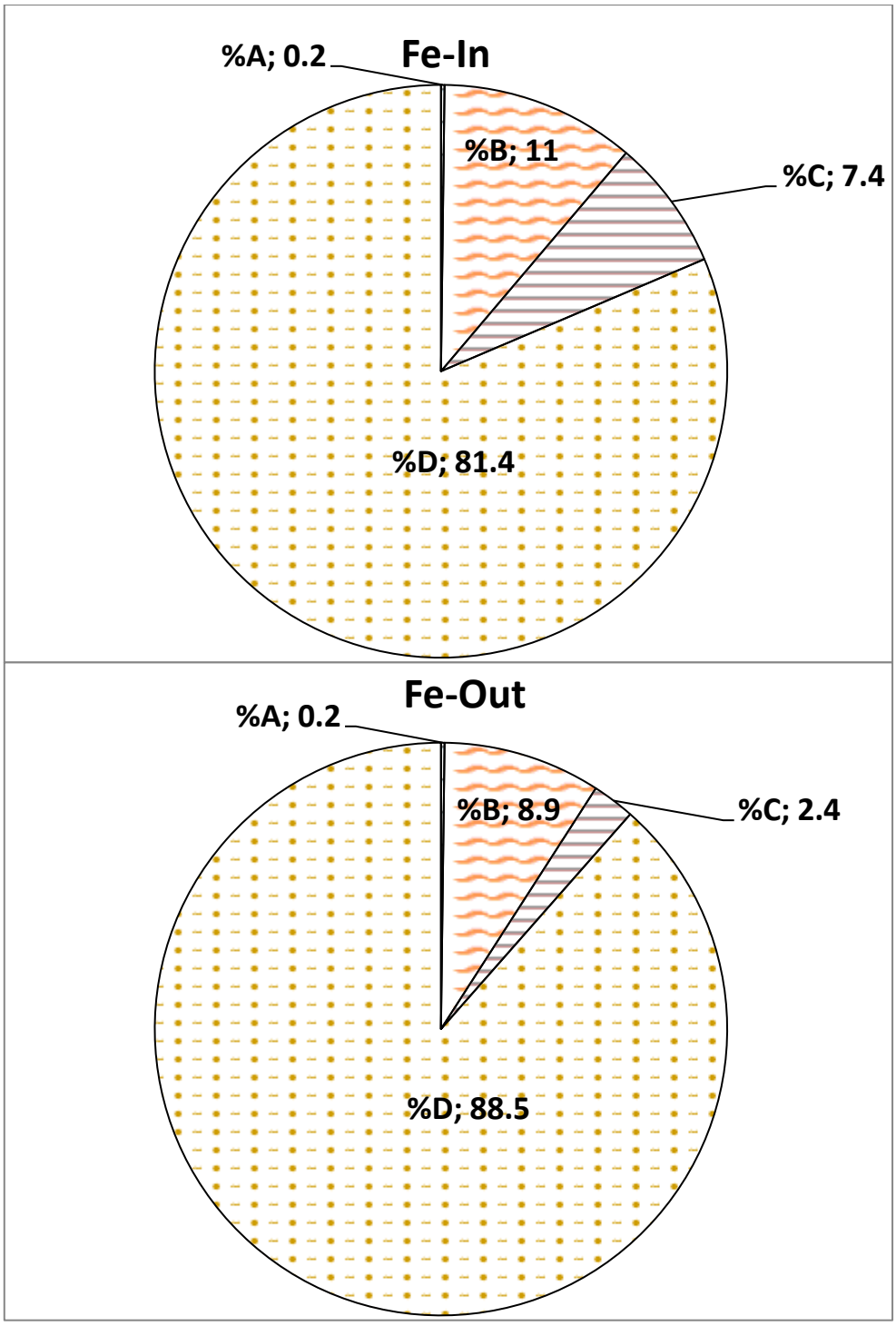


Figure 3-8: Percent distribution of Fe in various geochemical fractions to total metal concentration in inside and outside soil samples.

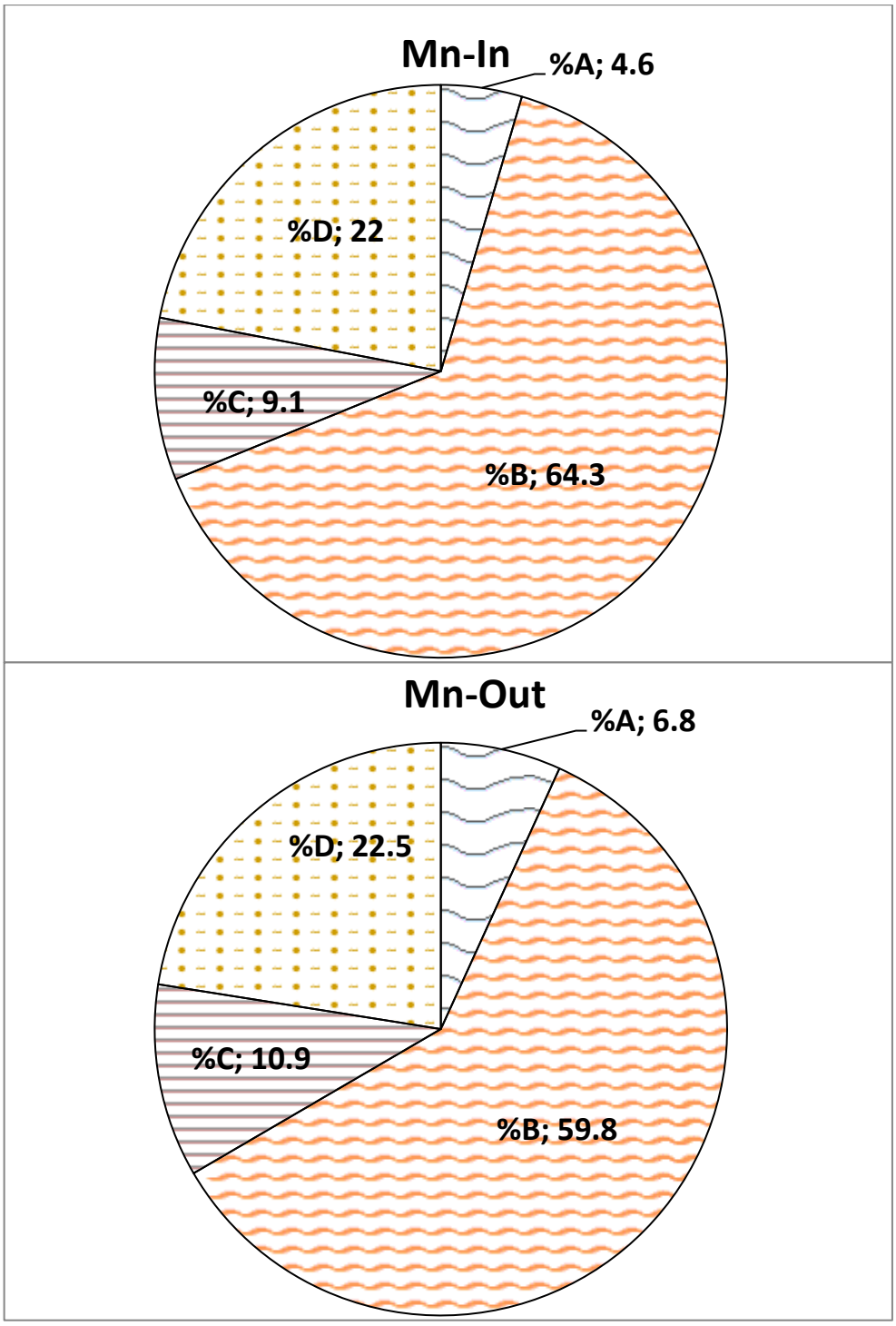


Figure 3-9: Percent distribution of Mn in various geochemical fractions to total metal concentration in inside and outside soil samples.

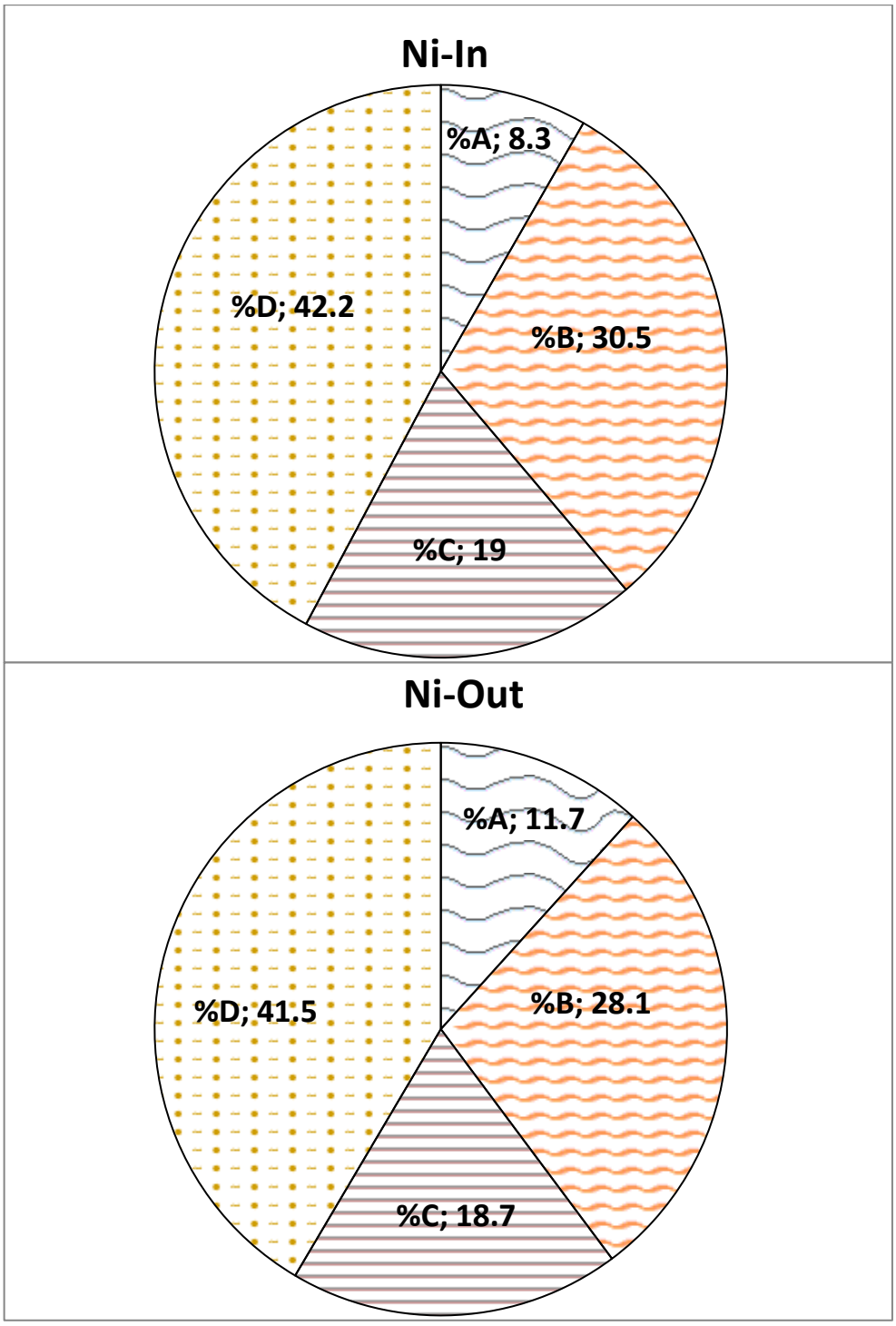


Figure 3-10: Percent distribution of Ni in various geochemical fractions to total metal concentration in inside and outside soil samples.

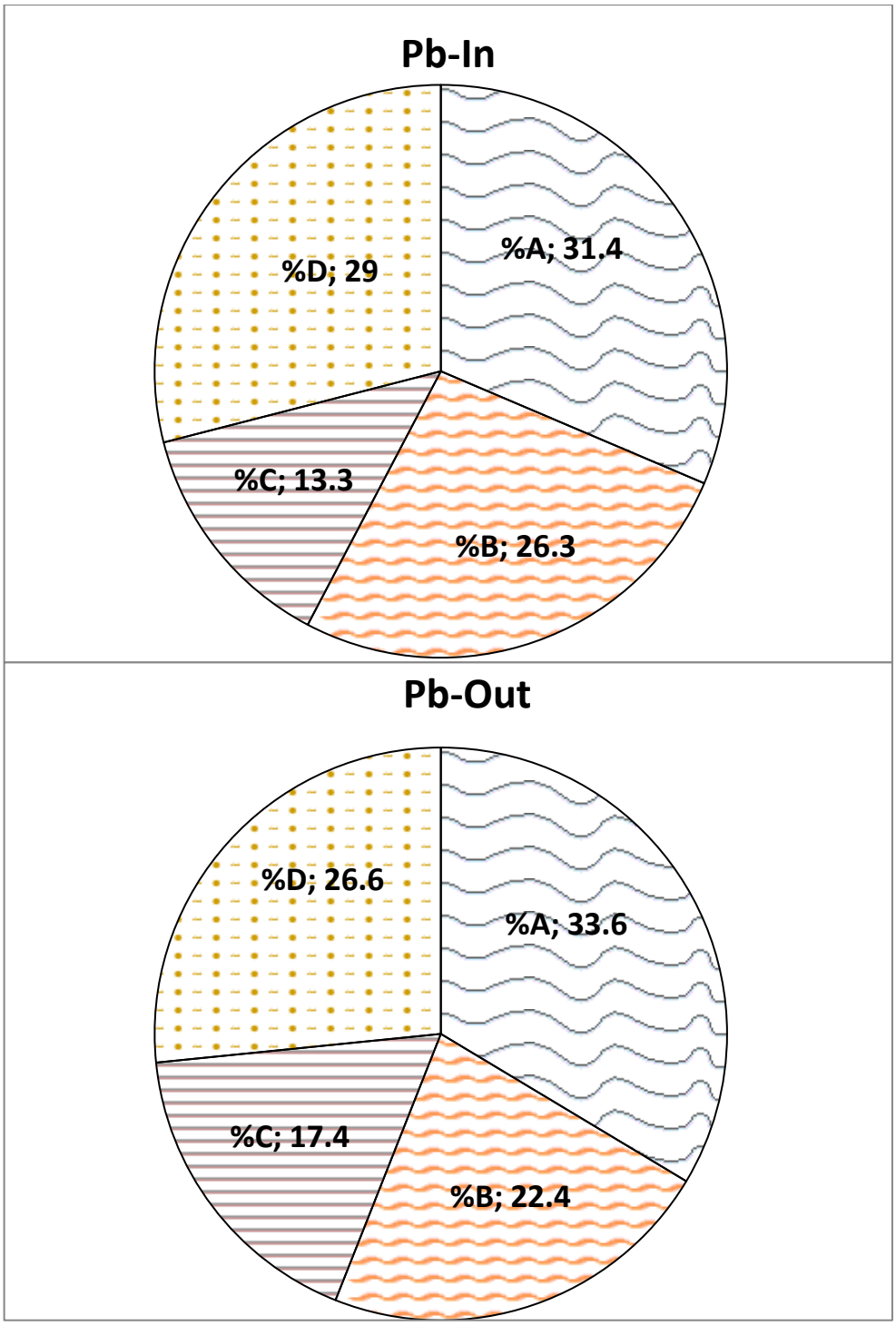


Figure 3-11: Percent distribution of Pb in various geochemical fractions to total metal concentration in inside and outside soil samples.

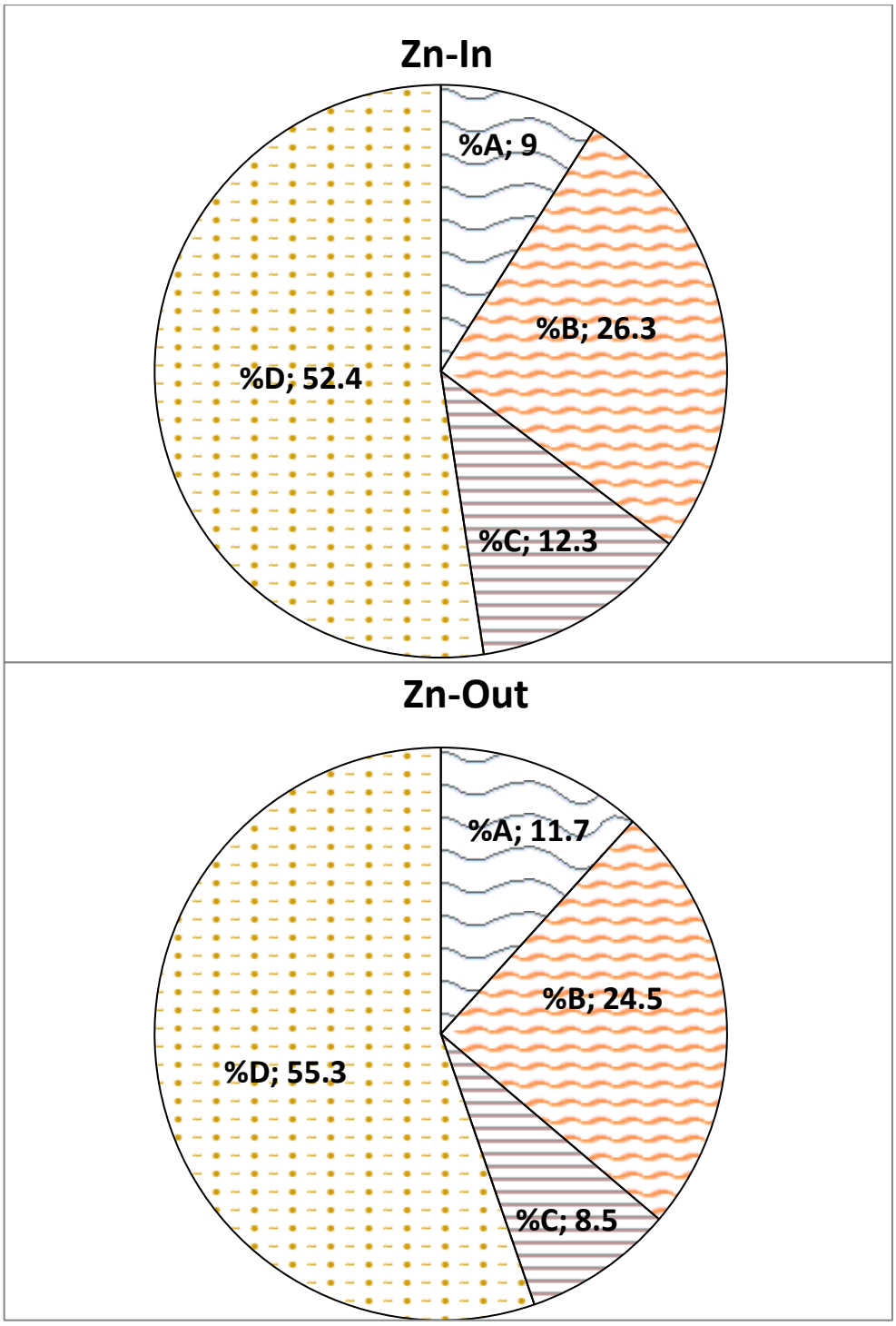


Figure 3-12: Percent distribution of Zn in various geochemical fractions to total metal concentration in inside and outside soil samples.

3.4. Concentration versus Sampling Direction

The influence of sampling directions on the observed levels of measured elements was investigated in this study. All samples from the same direction were averaged for each element and presented in Table 3.8 for the out-side samples only. In addition, the percent contribution by direction for each element were calculated and presented in Table 3.9.

Results in tables 3.8 and 3.9 indicate that samples collected from the western direction have lower levels than all other samples. However, except the western direction, levels of measured elements in soils from the other three directions seem to be similarly distributed, indicating that non of the directions is predominant.

Table 3.7. Average concentrations of heavy elements in samples from the four different directions.

Element	North	South	West	East
Cd	2.8±3.8	4.4 ± 1.1	2.2 ± 4.3	3.0 ± 6.1
Co	82.8 ± 29.6	98.7 ± 30.7	76.3 ± 36.8	86.8 ± 37.7
Cr	52.8 ± 27.6	69.8 ± 7.0	40.9 ± 43.8	56.5 ± 42.4
Cu	31.5 ± 1.0	32.0 ± 1.8	27.9 ± 3.3	34.0 ± 4.5
Fe	16075.6 ± 5288.6	17679.3 ± 6397.2	14437.1 ± 6095.5	16104.0 ± 5852.9
Mn	1008.2 ± 195.4	1536.4 ± 673.5	1008.0 ± 471.1	1206.1 ± 246.9
Ni	87.9 ± 24.8	109.8 ± 28.5	83.4 ± 31.9	99.2 ± 23.6
Pb	337.8 ± 181.1	389.0 ± 75.1	241.2 ± 55.4	320.8 ± 91.7
Zn	228.5 ± 105.2	271.5 ± 74.8	147.5 ± 39.2	173.2 ± 51.1

Table 3. 8. Percent of Average concentrations in samples from the four different directions.

Element%	North	South	West	East	Total %
Cd%	22.24	35.58	18.01	24.17	100
Co%	24.03	28.64	22.14	25.19	100
Cr%	23.99	31.73	18.58	25.7	100
Cu%	25.17	25.61	22.03	27.18	100
Fe%	25	27.5	22.45	25.05	100
Mn%	21.19	32.29	21.18	25.34	100
Ni%	23.12	28.86	21.93	26.09	100
Pb%	26.21	30.18	18.72	24.89	100
Zn%	27.84	33.08	17.97	21.1	100

3.5. Conclusions

Based on the results of the current study, the following points have been concluded;

1. Concentrations of measured elements are higher in soil samples collected from the close proximity of the factory (inside) than those collected far away from the factory (outside). This may roughly indicate that the concentrations of elements decrease as the distance from the factory increase.
2. The highest average concentrations were found in the ash samples for all elements.
3. There were significant differences between levels of Fe, Co and Mn in the inside soil samples as compared to the outside samples at 95% confidence levels.
4. Most of the Fe, Cr, Co, Ni, Cu and Zn are associated with the residual fraction.
5. Most of the Pb is associated with the bioavailable fraction.
6. Most of the Cd and Mn is associated with the oxides phase.
7. Lowest concentrations were observed in samples collected from the western direction.

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الملخص

الشرادقة، عماد خالد ، الاستخلاص التسلسلي للعناصر الثقيلة للتربة في محيط مصنع لصهر الحديد في المفرق ، الأردن . رسالة ماجستير في العلوم ، قسم الكيمياء ، جامعة اليرموك ، ٢٠١٥ ، (المشرف: الأستاذ الدكتور إدريس المومني).

تهدف هذه الدراسة إلى تقدير مستويات المعادن الثقيلة المختارة (الكادميوم ، الكوبالت ، الكروم ، النحاس ، الحديد ، المنغنيز ، النيكل ، الرصاص ، الزنك) في التربة حول واحد من مصانع صهر الحديد في الأردن (المصنع الصيني لصهر الحديد) في المفرق، تم جمع ٣٢ عينة من التربة السطحية داخل و خارج المصنع على عمق ٠-١٥سم و أربعة عينات من رماد المداخن خلال شهري نيسان ٢٠١٤ و أب ٢٠١٥ تم سحق العينات وغربلتها (٢٥٠ميكرون) ثم هضمت كليا باستخدام الأحماض القوية من خلال طريقة (US-EPA 3052)، أيضا تم استخلاص المعادن المستهدفة تسلسليا باستخدام طريقة (BCR). و كانت مراحل الاستخلاص أربعة أجزاء (١) ذائب في الأحماض الضعيفة وسهل التحرر (المتوفر بيولوجيا)(٢) قابل للاختزال (٣) قابل للتأكسد (٤) المتبقي (صعب التحرر).

تم تحليل جميع العينات باستخدام جهاز طيف الامتصاص الذري (FAAS) حيث تم الكشف عن تركيز العناصر التسعة التالية: Cd, Co, Cr, Cu, Fe, Mn, Ni, Pb, Zn ، و كان متوسط التركيز لهذه العناصر في عينات التربة التي جمعت من داخل المصنع على النحو التالي : ١٩٠٧٣ ± ٤٣٠١، ٨٦٧ ± ٢٤٠، ٥١٤ ± ٤٠٧، ٥٠٠ ± ٢٥١، ٩٨,٨٣ ± ١٨,٩٩، ٨٧,٩ ± ٢٢,٦، ٤٥,٥٨ ± ٥٩,٦٢، ٣٥,٢٩ ± ١١,٠٤، ٢,٧٢ ± ٣,٧٢ mg/kg . على التوالي

أما تركيز العناصر لعينات التربة من خارج المصنع فكانت كالتالي : ١٤٥١٦ ± ٥٥٣٩ ، ١١٤٢ ± ٣٩٨، ٢٧١ ± ١٧٢، ١٩٤ ± ٧٥، ٨٨,٦ ± ٢٧,٧، ٧٨,٢٦ ± ٣٢,٠٣ ، ٤٧,٥٤ ± ٣٣,١٨، ٣١,٨٨ ± ٤,٢٤، ٢,٥٨± ٤,٠٣ ، mg/Kg . على التوالي .

و كانت تركيز العناصر في عينات الرماد: ٢٢٢٢٠ ±٢٠٧٩، ٧٥٠٩± ١٥٤٠، ٢٩٨٦ ± ٩٥٨، ٢٩٥٥ ± ١٣٢، ٣٨٢,٠٦ ± ١٦١,٣٢ ، ٣٠٩,٧٤ ± ١٥٥,٦، ١٥٣,٧ ± ٨٥,١٢، ١١١,١ ± ٣٢,٤٢، ١٠٩,١٨ ± ٦٧,٧١ mg/kg على التوالي .

و وجد أن عناصر الكادميوم و الرصاص و الكوبالت تحتوي على اكبر نسبه من الجزء الاول (الذائب في الأحماض الضعيفة) بواقع ٤٩%، ٣٢% ، ١٦ % ، على التوالي . بينما كانت عناصر الكروم والحديد و النحاس و الزنك تحتوي على اكبر نسبة من الجزء الرابع (صعب التحرر) بواقع ٩٣% ، ٨٨%، ٧٥%، و ٥٥% . على التوالي .

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